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Implementation of a Non-coupled Integrated Transformer for a Two-phase Interleaved Flyback PFC Rectifier

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Abstract

Universal serial bus power delivery (USB PD 3.1) is a widely-applied standard for a lot of electronic device charging. In 2021, USB PD 3.1 was announced with a variable output of 5 V to 48 V and a maximum output power of 240 W. A multi-phase flyback rectifier is capable of power factor correction, galvanic isolation, and wide output voltage range, and is proposed as an alternative topology for USB PD. However, the number of magnetic components increases with the number of phases, which can lead to lower power density. This paper proposes the design of a non-coupled integrated transformer for a two-phase interleaved flyback rectifier to reduce the size and number of magnetic components. The independent operation of the integrated transformer in two phases was verified using a 240-W two-phase interleaved flyback PFC rectifier at 100 kHz switching frequency operating at discontinuous conduction mode, both in simulation and experiment. After conducting a volume analysis through the area product method, the proposed transformer is up to 19% smaller compared to its conventional counterpart.

Keywords: Integrated magnetics, Transformer design, two-phase interleaved flyback rectifier, USB PD 3.1, PFC, Core volume analysis

Contents

1 Introduction **1**

1.1 Power Electronics 1

1.2 USB Power Delivery 3

1.3 Topologies for implementing USB Power Delivery 3.1 4

1.3.1 PFC Boost, LLC, and Buck Converter Combination Circuit . 4

1.3.2 PFC Boost with AHB Flyback 5

1.3.3 Conventional PFC Flyback 6

1.3.4 Two-phase Interleaved Flyback PFC Rectifier 7

1.4 Concept of an Integrated Transformer for the Two-phase Interleaved Flyback PFC Rectifier 9

1.5 Objectives of the Study 10

1.6 Signficance of the Study 11

1.7 Flow of the Paper 11

2 Weak Form Formulation **13**

2.1 Integrated Magnetics 13

2.2 Common Integrated Magnetics Techniques 14

2.2.1 Integrated Transformer by Ishihara et. al. (2015) 15

2.2.2 Integrated Transformer by Taheri et. al. (2011) 17

2.2.3 Integrated Transformer by Qian and Lehman (2007) 19

2.2.4 Integrated Magnetic Component by Yan et. al. (2003) 21

2.3 Proposed Integrated Transformer 23

3 Problem Setup **24**

3.1 Circuit and Transformer Operational Analysis 27

3.2 Saturation Magnetic Flux Density Analysis 38

3.3 Miniaturization Effect of the Proposed Transformer 45

4 Results and Discussion **51**

4.1 Turns Ratio 51

4.2 Saturation Magnetic Flux Density 52

4.3 Magnetizing Inductance 54

4.4 Number of Turns for the Transformers 55

4.5 Air Gap Design 57

5 Summary and Conclusion **61**

Bibliography **64**

List of Figures

1.1 An example of power delivery technology for a laptop charger. The charger on the right is very small compared to the conventional one on the left, due to PD, and is connected to the PC via USB-C. 3
1.2 PFC boost, LLC, and Buck converter combination circuit in [6], [7]. . 5
1.3 PFC boost with AHB flyback in [5], [8]. 6
1.4 Conventional boost PFC flyback converter. 7
1.5 A two-phase interleaved flyback PFC rectifier. 8
1.6 Conventional transformer of the two-phase interleaved flyback PFC rectifier. 9
1.7 Two-phase interleaved flyback PFC rectifier with the proposed integrated transformer. 10

2.1 The proposed integrated transformer in [16]. 15
2.2 The proposed integrated transformer in [18]. 17
2.3 The proposed integrated transformer in [19]. 19
2.4 The proposed integrated magnetic component in [20]. 21
2.5 The proposed non-coupled integrated transformer in this study. . . . 23

3.1 The proposed two-phase interleaved flyback PFC rectifier. 24
3.2 The proposed non-coupled integrated transformer in this study. . . . 25
3.3 Flux densities inside the proposed integrated transformer. B_l represents the flux density in the left leg, B_r represents the flux density in the right leg, and B_c represents the flux density in the right leg. . . . 26
3.4 Integrated transformer separated according to AC operation cycle. . . 26

3.5 Key electrical and flux density waveforms of the circuit when $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$. 29
3.6 Operation of the proposed circuit with the transformer at Mode 1, $t_1 < t < t_2$. 30
3.7 Operation of the proposed circuit with the transformer at Mode 2, $t_2 < t < t_3$. 31
3.8 Operation of the proposed circuit with the transformer at Mode 3, $t_3 < t < t_4$. 32
3.9 Operation of the proposed circuit with the transformer at Mode 4, $t_4 < t < t_5$. 33
3.10 Operation of the proposed circuit with the transformer at Mode 5, $t_5 < t < t_6$. 34
3.11 Operation of the proposed circuit with the transformer at Mode 6, $t_6 < t < t_7$. 35
3.12 Equivalent magnetic circuits for the proposed transformer. 37
3.13 Magnetic flux density waveforms for Case 1: $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$. 39
3.14 Magnetic flux density waveforms for Case 2: $D \leq 0.5$ and $D > 0.5\frac{nV_O}{V_{\text{in, max}}}$. 40
3.15 Magnetic flux density waveforms for Case 3: $D > 0.5$. 43
3.16 EE-core with a simplified A_w and A_c [23]. 46
3.17 Representation of the conventional transformer with windings. 47
3.18 Representation of the proposed transformer with windings. 48

4.1 Geometric representation of the transformer. Here, the effective cross-sectional area is given as A_{mo} for the outer legs and A_{mc} for the center leg. The effective magnetic path length is given as l_{mo} for the outer legs and l_{mc} for the center leg. 55
4.2 Magnetic resistances for the proposed transformer. Here, the magnetic resistance is given as R_{mo} for the outer legs and R_{mc} for the center leg. 57

4.3 Equivalent magnetic circuits for the proposed transformer. 58
4.4 The air gaps of the proposed transformer, with $l_{mo_{air}} > l_{mc_{air}}$. 59
4.5 The air gaps of the proposed transformer, with $l_{mc_{air}} > l_{mo_{air}}$. 60

List of Tables

1.1 USB power delivery standards [4]. 3

3.1 Parameters used in the volume analysis. 46

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Chapter 1

Introduction

1.1 Power Electronics

Power Electronics is a discipline of Electrical Engineering that deals with conversion of power from one form to another. The main goal of power electronics is to promote power conversion without significant loss of energy, ensuring high efficiency, reliability, and compactness of the power conversion systems [1], [2].

A converter is an electrical circuit designed to change an input voltage into a different output voltage. Depending on the application, both the voltage level and the kind of voltage being outputted (i.e. DC or AC) may be manipulated [3].

There are four common kinds of converters commonly implemented today. These include: DC-DC Converters, AC-DC Converters (Rectifiers), DC-AC Converters (Inverters), and AC-AC Converters[3].

DC-DC Converters. DC-DC Converters convert a DC input to a DC output, and they can either raise up the voltage level (boost type or step-up DC-DC Converter), or bring down a voltage level (buck type or step-down DC-DC Converter). A buck converter is commonly used in various appliances that we own today. For instance, for charging, once AC is converted to DC for the battery, some components for our appliances need lower level DC to operate safely. Buck converters can be used here. LED Lighting systems commonly use Buck Converters to run efficiently.

It is also used a lot in automotive applications, especially for powering small electrical components that require only a small amount of voltage to operate. A boost converter is also used commonly in battery-powered devices. For instance, some components in our automobiles, or laptops, or smartphones require higher voltage to operate.

DC-AC Converters. DC-AC Converters convert a DC input to an AC output of desired voltage and frequency level and are commonly known as inverters. They commonly utilize PWM Control to be able to convert voltage. Their most common applications include the automotive industry, especially for powering air conditioners, and electric motors for electric vehicles. During power outages or camping excursion for example, portable generators are often used to power various appliances, and in order to do so, they convert DC to AC first, which then go into the appliances for their own conversion to DC and finally to be used. Lastly, they are also used a frequently in industrial applications, especially for controlling motors, and in electro-welding and electro-forming processes.

AC-DC Converters. AC-DC Converters convert an AC input to a DC output, and are also commonly known as rectifiers. These are commonly used in charging systems and power supplies too, as had been explained, and they take care of the part of converting AC into usable DC by our devices. It is also commonly used in some renewable power systems, such as solar power systems, to convert AC from the solar panel to DC power stored in batteries, which can then be used to power various devices.

AC-AC Converters. AC-AC Converters convert an AC input to an AC input (to a different level) also. One of the most common applications is power distribution from an electrical grid. Power lines from power plants carry very high level AC, and for it to be usable in a home setting, this AC must be converted to a different level and frequency for it to be usable in our appliances.

This study focuses on the circuit for USB Power Delivery, which is an example

of such application is electric charging, which makes use of power conversion from power supply to a load. More specifically, it is an AC-DC converter used in a lot of applications.

1.2 USB Power Delivery

USB Power Delivery (PD) is a widely-applied charging technique today. It is widely used to promote fast charging for instance for laptops, devices, and other everyday household appliances at a very small and powerful package. An example of the extent of miniaturization of USB PD is shown in Figure 1.1. Laptop chargers, for instance, are becoming smaller and are able to deliver higher power with a small package due to the use of PD. A summary of the specifications of recent PD Standards are summarized in Table 1.1.

Figure 1.1: An example of power delivery technology for a laptop charger. The charger on the right is very small compared to the conventional one on the left, due to PD, and is connected to the PC via USB-C.

Table 1.1: USB power delivery standards [4].

USB power supply standard	Maximum supply current	Supply voltage	Maximum supply power
USB	0.9 A	5 V	4.5 W
USB BC 1.2	1.5 A	5 V	7.5 W
Type-C	3 A	5 V	15 W
USB PD 3.0	5 A	5 - 20 V	100 W
USB PD 3.1	**5 A**	**5 - 48 V**	**240 W**

The latest standard is called USB PD 3.1, which allows for up to 240 W of output power across a wide output voltage range. To implement USB PD 3.1, the following specifications are required [4], [5]:

- **PFC Capability:** be capable of power factor correction (PFC) for supporting 75 W and above, for improved efficiency and less losses.
- **User Safety:** be equipped with galvanic isolation for user safety.
- **Wide Output Voltage Range**: be capable of providing an output voltage of 5 V to 48 V.

With the newly announced standard that emphasizes higher power applications, various changes will be necessitated for existing devices today. More specifically, the development of new topologies while maintaining high power density, efficiency, and low cost are new challenges faced today [5].

1.3 Topologies for implementing USB Power Delivery 3.1

There are a lot of different topologies available today to implement USB Power Delivery. This section will introduce some of their advantages and disadvantages.

1.3.1 PFC Boost, LLC, and Buck Converter Combination Circuit

A combination of a PFC Boost, LLC, and Buck Converter was introduced in [6], [7]. It is expected to be applied for laptop power adaptors. The circuit diagram for the topology is shown in Figure 1.2. The topology allows for high power density and efficiency. However, the voltage output range is very small (16.5 V - 18.5 V) and the power output does not cover up to 240 W, and is thus not flexible for a lot of applications and cannot meet the standard. Moreover, there are going to be

three stages of conversion, as a buck converter is still needed after the conversion by the topology shown in the figure. This then entails that this circuit is going to be more expensive to manufacture, and moreover, with the number of components, the circuit is going to be more difficult to implement.

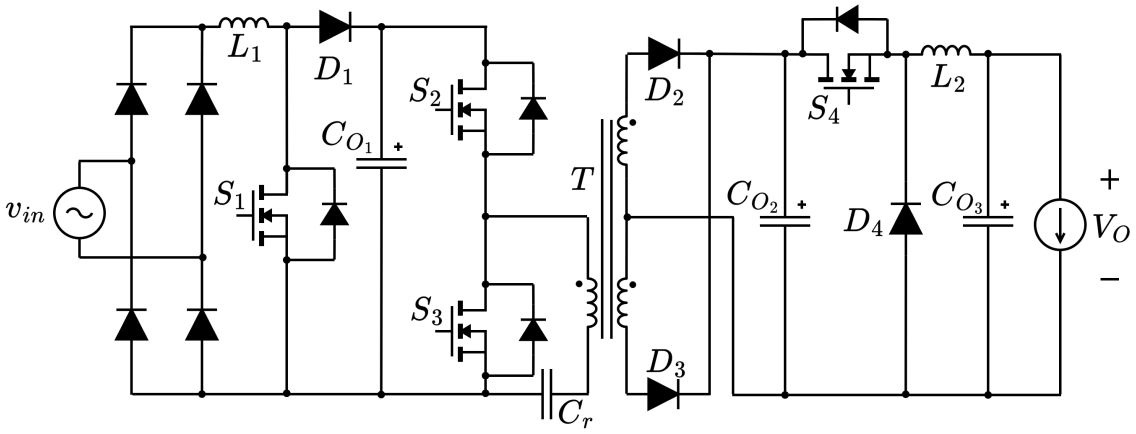


Figure 1.2: PFC boost, LLC, and Buck converter combination circuit in [6], [7].

1.3.2 PFC Boost with AHB Flyback

A PFC Boost circuit that follows an Asymmetrical Half-bridge Flyback Rectifier, also known as a Hybrid Flyback, was introduced in [5], [8]. The circuit diagram for the topology is shown in Figure 1.3. It implements the Zero-Voltage Switching (ZVS) technique to improve efficiency and power density of the converter overall. It also follows the specifications of USB PD 3.1, with a wide input voltage range of up to 420 V and an output of $5 - 48$ V, up to a maximum output of 240 W. Although this topology is effective, reaching up to 98% efficiency, implementing ZVS requires a delicate consideration of the transformer leakage inductance, its magnetizing inductance, and also the resonant capacitance as well, making the design method of the transformer and also implementation and design of the circuit more complicated than usual.

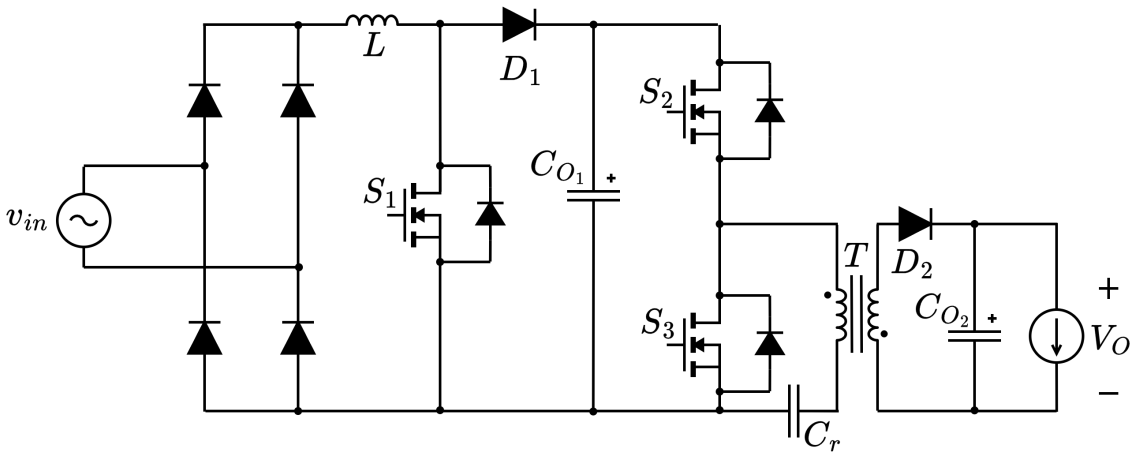


Figure 1.3: PFC boost with AHB flyback in [5], [8].

1.3.3 Conventional PFC Flyback

A conventional PFC Flyback is easy to implement and the transformer is easy to design as well. A basic flyback PFC rectifier is shown in 1.4. Moreover, there are a lot of intricacies regarding the conventional flyback, such as the imminent noise on the switching waveforms due to the resonance between the flyback transformer and the current ripple in the input and output side as well. To improve efficiency, a few other improvements to the conventional topology has been proposed, such as clamped forward active flyback converters [9], [10] and also bridgeless flyback converters [11], [12], [13], [14].

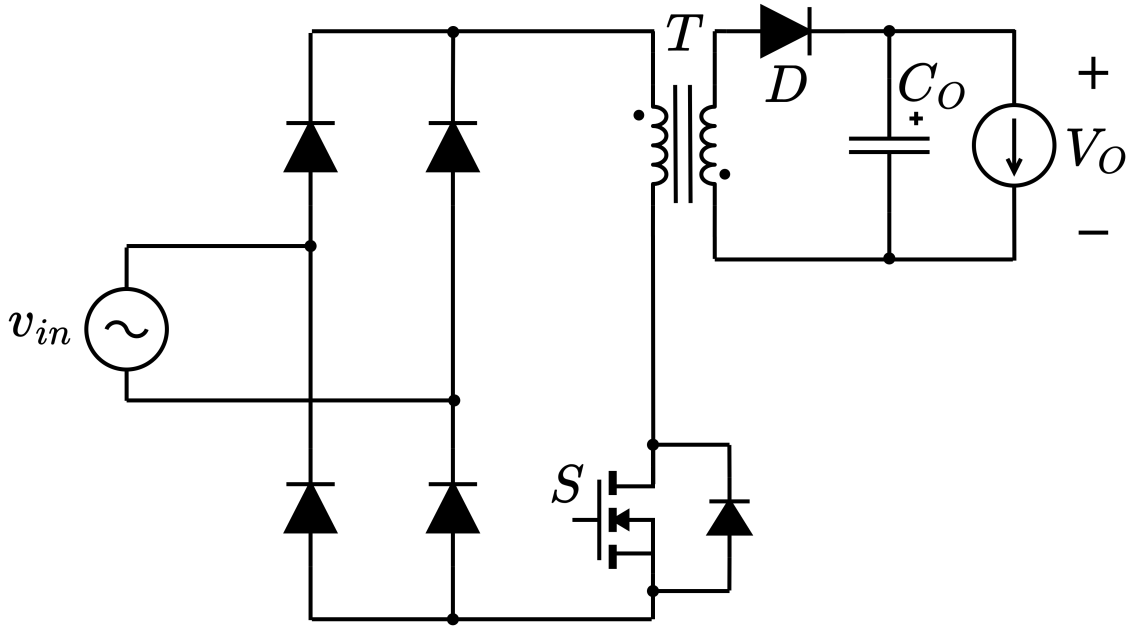


Figure 1.4: Conventional boost PFC flyback converter.

Although the conventional is very easy to design, due to the current stress on the switches especially at higher powers, one-phase is not enough to accommodate all the conversion due to the high current on switches. The flyback converters mentioned formerly were used only for around 100 W, and are thus more practical for lower power applications [9], [10], [11], [12], [13], [14].

1.3.4 Two-phase Interleaved Flyback PFC Rectifier

The circuits examined have their own benefits, yet some do not meet PD 3.1 specifications, are complex to design, and are costly due to circuit complexity. To solve these problems, interleaving is proposed. A two-phase interleaved flyback PFC rectifier is examined in this study, as shown in Figure 1.5, with the circuit operating in DCM, and driven at two phases with 180° phase difference. With the inputs in parallel, one main benefit of interleaving is to reduce current stresses on the switches, since half of the total power delivered to the load is transferred from the primary side to the secondary side in each phase. The input and output current ripples

are also expected to be reduced with interleaving. This then allows the circuit to accommodate higher load conversion [15], [16]. Moreover, from Figure 1.5, since the source nodes of switches S_1 and S_2 are common, only one power supply is required for the gate drive circuits. A common cathode package can be used for diodes D_1 and D_2. Therefore, the number of components is lesser than the circuits that entail a combination of PFC Boost, LLC, and Flyback, and also the hybrid AHB Flyback. The operating principle of this circuit will be discussed in detail in Chapter 3.

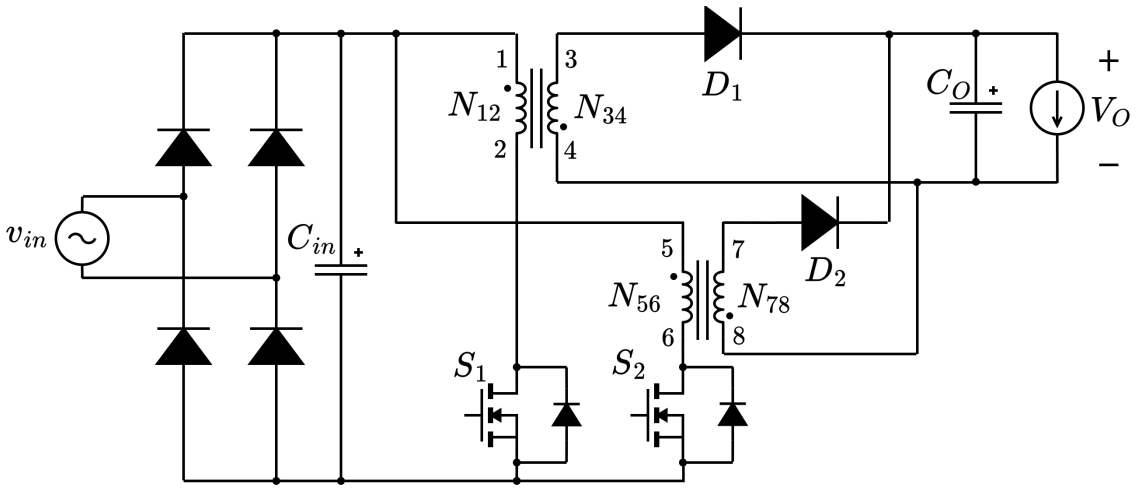


Figure 1.5: A two-phase interleaved flyback PFC rectifier.

However, a problem remains from this circuit. From figure 1.5, the circuit uses two separate magnetic elements, as shown in Figure 1.6. An increase in the number of phases means an increase in the volume of magnetic components, which implies lower power density. Moreover, magnetic cores are made of rare earth metals, such as nickel, manganese, and cobalt, and various base metals, such as iron, aluminum, and copper, and with increasing economic growth, their cost is expected to increase [16], [17]. This will also significantly increase the manufacturing cost of the circuit. Various USB PD 3.1 adapters should remain small and affordable for the public, and thus this issue must be addressed.

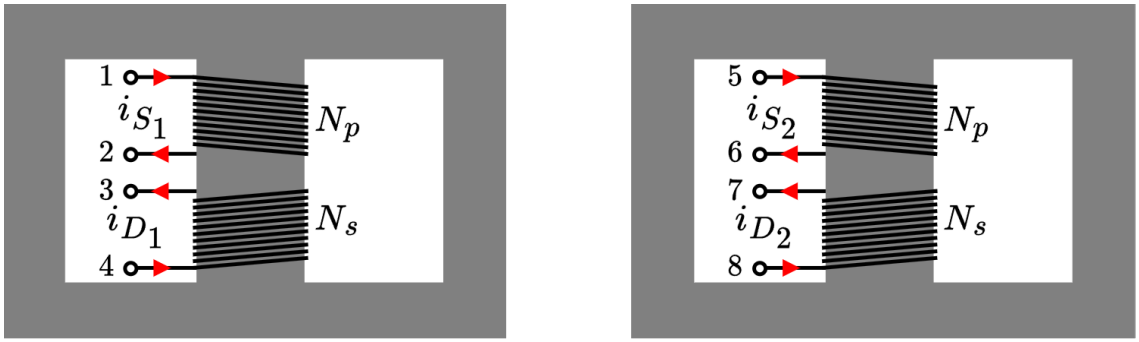
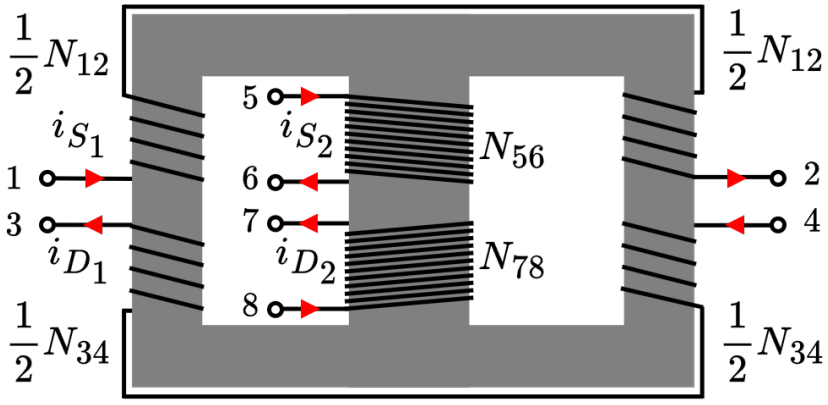


Figure 1.6: Conventional transformer of the two-phase interleaved flyback PFC rectifier.

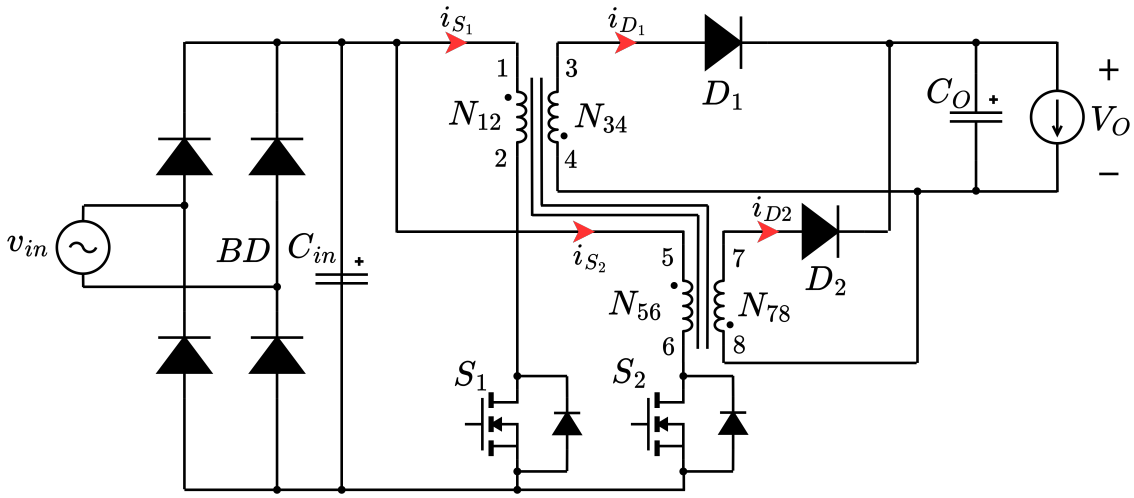
An integrated transformer is proposed to solve the problem and contribute to the miniaturization of the circuit. The proposed transformer can operate separately in two phases due to the complete elimination of coupling between the two transformers.

1.4 Concept of an Integrated Transformer for the Two-phase Interleaved Flyback PFC Rectifier

The miniaturization of the circuit is accomplished by a non-coupled integrated transformer that can operate separately in the two phases of the circuit operation. The proposed transformer is shown in Figure 1.7a and the circuit diagram with the proposed transformer is shown in Figure 1.7b. This is accomplished by the winding structure for the transformer shown in Figure 1.7a and is explained further in Chapter 3. The number of turns between terminals i and j corresponds to N_{ij} in Figure 1.7b. The winding of the proposed integrated magnetic component employs zero coupling coefficient between the two transformers $N_{12}N_{34}$ and $N_{56}N_{78}$. More details about the operational principle of the transformer will be shown in Chapter 3.



(a) Proposed non-coupled integrated transformer.



(b) Two-phase interleaved flyback PFC rectifier with the proposed transformer.

Figure 1.7: Two-phase interleaved flyback PFC rectifier with the proposed integrated transformer.

Integrated magnetics have been a widely-used method for magnetic component miniaturization, and are specially designed based on the circuit topology and purpose of the user. A review of some techniques are shown in Chapter 2.

1.5 Objectives of the Study

The study has three main goals. Firstly, this study aims to propose a non-coupled integrated transformer that can act separately for a two-phase interleaved flyback PFC rectifier operating in DCM. Important parameters, such as the saturation magnetic flux density and peak currents, are examined to properly select a magnetic core

and electronic components for the circuit, and is accomplished by a detailed theoretical operational analysis of the circuit and transformer. Secondly, this study aims to confirm the feasibility of the proposed transformer in simulation and experiment up to 240 W. Independent, two-phase operation between the two flyback converters in Figure 1.7a are confirmed in both simulation and experiment. Feasibility of the circuit to operate in one-phase drive is also confirmed. Lastly, this study aims to examine the extent of miniaturization offered by the proposed integrated transformer. A detailed volume analysis is also performed in this study. The efficiency of the circuit, both in simulation and experiment, is presented.

1.6 Signficance of the Study

This study is expected to contribute to the existing field of integrated magnetics. Moreover, the method presented here is one of the few many ways to promote circuit miniaturization and cost reduction, and thus this study is expected to contribute to the many ways of cost reduction and circuit miniaturization, especially for USB PD 3.1. Moreover, as will be explained in Chapter 2, most integrated magnetics do not employ zero coupling between the two seperate transformers, and thus, this study is expected to present a simpler alternative to existing techniques.

1.7 Flow of the Paper

The flow of the paper is as follows.

A short introduction to various existing integrated magnetics techniques for circuits that are similar to the two-phase interleaved flyback PFC rectifier in this study is shown in Chapter 2. The main difference and advantage of the proposed transformer is also detailed.

A detailed operational analysis of the electrical and magnetic characteristics of the circuit and the proposed transformer is explained in Chapter 3. Important design parameters, such as the peak current and saturation magnetic flux density

are also presented in Chapter 3. More details on the proposed winding structure is also presented. Lastly, an insight on the extent of miniaturization is also presented with a detailed volume analysis using the Area Product Method.

The design method of the proposed transformer is explained in Chapter 4. The validity of the design method is verified in Chapter 5, and examples are presented as well, both in simulation and experiment. The effect of coupling on circuit and operation and various explanations on the efficiency of the converter with the proposed transformer are also presented.

Conclusions and recommendations from the present status of work is presented in Chapter 6.

Chapter 2

An Overview of Integrated Magnetics

2.1 Integrated Magnetics

Integrated magnetics are a long existing technique for miniaturization of the magnetic components in a circuit. This section will detail some existing techniques that are available for integrated magnetics. The goal is to reduce the number of magnetic cores, as they are the component that can take up the most volume in terms of passive components. To be able to integrate multiple components into one, the winding structure should be carefully thought of.

To design an integrated magnetic component, one must take into account the following:

- **Circuit operational principle.** Circuits operate differently from each other. Some circuits may operate in for example, DCM or CCM. Some circuits may operate in step-up or step-down operation. Some circuits operate in different number of phases. Depending on these conditions, the transformer will conduct consecutively in the primary and secondary side. The operation frequency is also important to take into account. The flux interaction caused by the winding structure must then be carefully taken into account during the

design process to realize the desired circuit operation.

- **Target feature or improvement.** Some magnetic components are designed to improve certain features in the circuit. For instance, some magnetic components are designed to improve current ripple in the input and output side. Some magnetic components are designed to improve efficiency by controlling the current ripple in various parts of the circuit. Some magnetic components are designed for minaturization of other elements in the circuit as well.

- **Flux operation.** The flux interaction in the transformer can affect the circuit behavior and the size requirement of the core used for magnetic integration. This should be carefully considered in design to optimize the core used for the transformer, and is again commonly dependent on the winding structure. Some utilize coupling or flux cancellation to affect overall flux in the core.

2.2 Common Integrated Magnetics Techniques

There is no integrated magnetic technique made exactly for the circuit being examined in this study. However, in this section, four studies are examined which employ an integrated transformer in two-phase driving. The focus of each study is different, and only some of the most significant contributions are detailed in this review. The difference of the proposed non-coupled integrated transformer compared to the other transformers or magnetic elements examined in this section is also detailed.

2.2.1 Integrated Transformer by Ishihara et. al. (2015)

The transformer was implemented in a two-phase interleaved forward converter, operating in continuous current mode (CCM), at a step-up and step-down operation. The proposed transformer in [16] is shown in Figure 2.1, which is the transformer implemented in parallel-input mode. The step-down operation will be reviewed in this section.

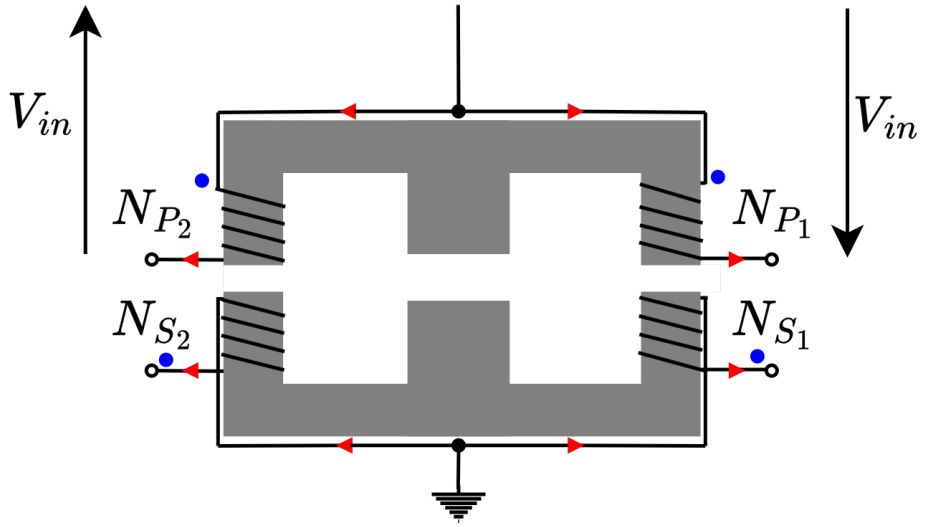


Figure 2.1: The proposed integrated transformer in [16].

The authors of [16] examine the transformer in both series and parallel input connection. The transformer was connected in parallel at the output side. If one wants to reduce the voltage stress on the switching elements, the inputs must be connected in series, whereas to reduce the current stress, the inputs must be connected in parallel.

The primary and secondary side windings are placed on the outer legs only. The winding is designed such that no flux cancellation occurs in any of the legs. An air gap is placed in the center leg. The transformer was designed for 400 V − 24 V conversion at 200 W power output. Magnetic saturation does not occur in all the legs of the transformer. The study identified the input capacitor as the component with the largest loss and the transformer in this study was focused to reduce the

size of the input capacitor. A thorough loss analysis was provided in this study. From this study, it was found that the series input configuration was more preferred more than the parallel configuration for the step-down condition, as it proved to have less losses and higher power density compared to the parallel configuration at the testing specifications. However, for this circuit and its transformer, the input voltage is very high compared to the input voltage used in this study. The loss reduction is not the focus of this study, and the proposed transformer in study has a different focus compared to the study described here.

The design method uses the primary side current ripple and the coupling coefficient to decide the saturation magnetic flux density, to be able to decide the appropriate core. This makes the design method more complicated than usual.

2.2.2 Integrated Transformer by Taheri et. al. (2011)

The transformer was implemented in a two-phase interleaved forward converter, operating in continuous current mode (CCM), at a step-down operation. The proposed transformer in [18] is shown in Figure 2.2. The proposed transformer and converter were not verified experimentally in this study. The transformer was connected in series in the input side and in parallel at the output side. This was done to reduce the voltage stress in the switching devices in the primary components. Moroever, because of the parallel connection in the secondary side, ripple current will be reduced in the output side. There is simultaneous conduction in the secondary side.

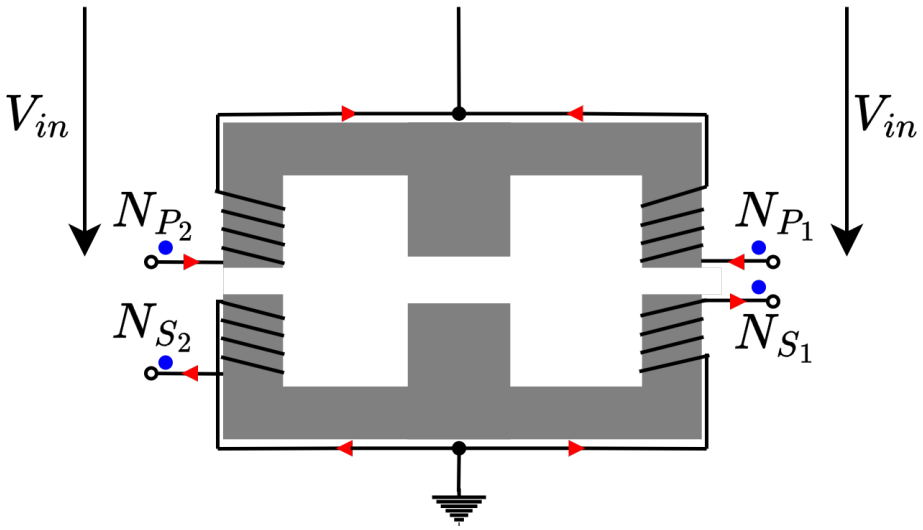


Figure 2.2: The proposed integrated transformer in [18].

The primary and secondary side windings are wound on the outer legs only, and the winding makes it possible for flux in the center leg to be cancelled in the center leg, which can therefore reduce the amount of flux in the core over all, and can therefore contribute to less core loss. The windings are also implemented such that there is coupling between the primary and secondary side windings, which can reduce the input current ripple, although the amount of coupling was not specified in the study, and moreover, the topology this transformer is being implemented into is different from the one focused in this study.

The circuit used in this study enables ZVS, which is a well-known technique to improve efficiency and various circuit characteristics, and this then makes the design of the transformer more complicated, as the values of the transformer leakage and magnetizing inductance should be chosen and set accordingly with the switch capacitance. Moreover, the design method uses the primary side current ripple and the coupling coefficient to decide the saturation magnetic flux density, to be able to decide the appropriate core. This makes the design method more complicated than usual.

2.2.3 Integrated Transformer by Qian and Lehman (2007)

The transformer was implemented in a two-phase interleaved flyback converter, operating in continuous current mode (CCM), at a step-down operation, for high input voltages. The transformer in [19] is shown in Figure 2.3. The transformer was connected in series in the input side and in parallel at the output side. This was done to reduce the voltage stress in the switching devices in the primary components. There is simultaneous conduction in the secondary side. A lot of switching devices already have a high voltage rating. However, the current stresses will not be reduced in the switching devices, which may necessitate switching devices with higher current ratings and are more expensive. The parallel connection in the secondary side can still keep the benefit of ripple current in the output side.

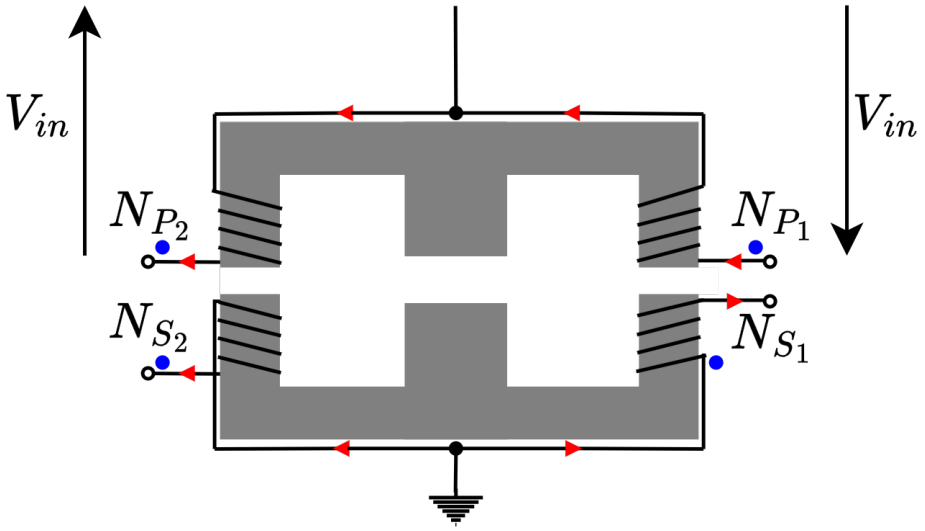


Figure 2.3: The proposed integrated transformer in [19].

The primary and secondary side of the transformers were inversely coupled with each other. Moreover, all the legs were gapped, causing a weak coupling between the windings, and therefore contributing to the reduction of current spike caused by voltage mismatch. However, this coupling causes a large leakage inductance, and can cause problems in distortion and noise in the waveforms, as can be seen in the results of this study. This is mainly due to the fact that flux can go through all the

legs at all times. These points can be improved, although is not the main focus of the proposed integrated transformer in this study.

The transformers are wound on the outer legs only, and the winding structure makes magnetic flux non-cancellable in the transformer at any time. The design method uses the primary side current ripple and the coupling coefficient to decide the saturation magnetic flux density, to be able to decide the proper core. This makes the design method more complicated than usual.

2.2.4 Integrated Magnetic Component by Yan et. al. (2003)

The integrated magnetic component in [20] was implemented in a full-wave converter, operating in continuous current mode (CCM), at a step-down operation. There is simulatneous conduction in the primary and secondary side. The proposed integrated magnetic component by the authors is shown in Figure 2.4. The transformer was connected in parallel at the input side and in parallel at the output side.

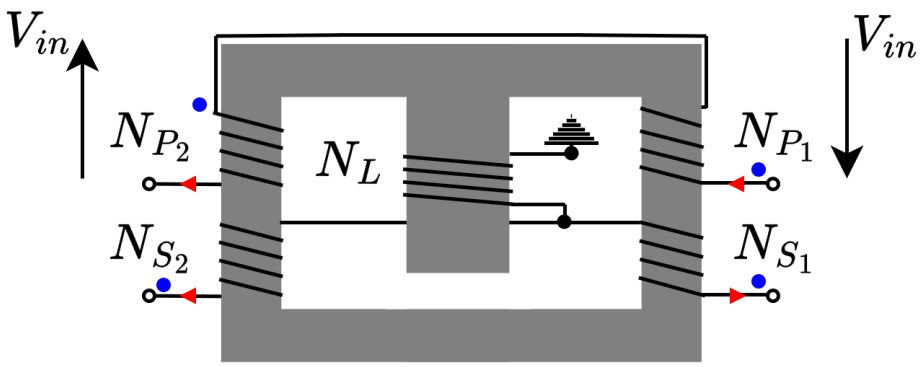


Figure 2.4: The proposed integrated magnetic component in [20].

Compared to the other integrated transformers introduced in this study, this transformer features an additional inductor winding at the center leg. The inductor winding was added to control the output ripple. Flux cancellation does not occur in the core also. Moreover, this transformer was implemented using a planar core, which can then improve flux distribution and help in magnetic element volume miniaturization. However, it is important to note that this integrated magnetic component was designed in low voltage specifications (75 V − 3.3 V conversion), and can only support up to 100 W. The efficiency of the circuit is considerable at 92.9%.

The design method uses the primary side current ripple and saturation magnetic flux density, to be able to decide the proper core. No information about the coupling coefficient is given and seems to be not a large consideration in the study, although

there seems to be coupling according to the results.

2.3 Proposed Integrated Transformer

Compared to other integrated magnetics techniques, the proposed integrated transformer in this study, shown in 2.5 realizes zero coupling coefficient between the two separate transformers. This then allows the user to operate the circuit in one-phase only if desired, which allows more flexibility in terms of circuit operation. For instance, in low-powers, it might be better to drive the circuit in single-phase instead to reduce switching losses and improve efficiency, which is especially important since USB PD operation ranges from low to high power. Moreover, driving the circuit without coupling between the two transformers makes it simpler to implement the circuit. Transformer design also becomes easier as the effect of mutual inductance is not taken into account in the various design parameters of the transformer design. Lastly, the design is also highly dependent on the duty cycle, which is a parameter that is easily decided for the circuit. An additional benefit of implementing a non-coupled integrated transformer is that the circuit will be easier to implement. Details of the winding and operational principle are shown in Chapter 3.

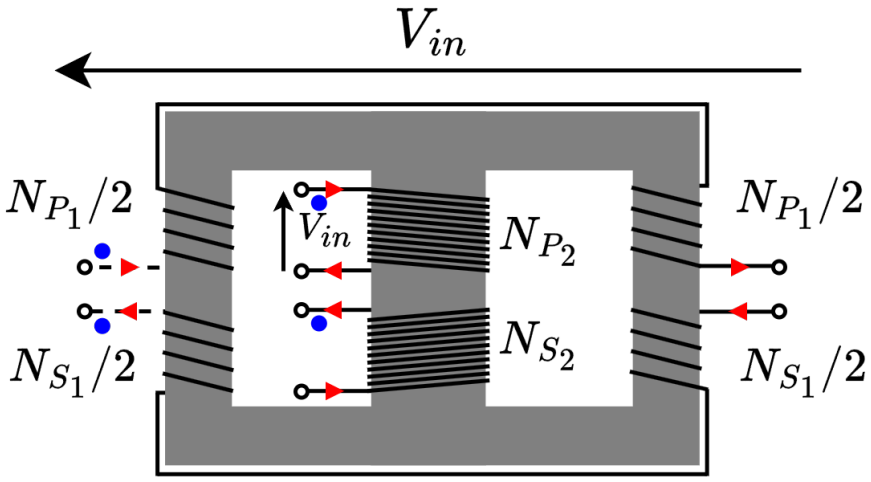


Figure 2.5: The proposed non-coupled integrated transformer in this study.

Chapter 3

Operational Analysis and Core Volume Analysis

This chapter will describe the operational principle of the two-phase interleaved PFC flyback rectifier analyzed in this study with the proposed integrated transformer.

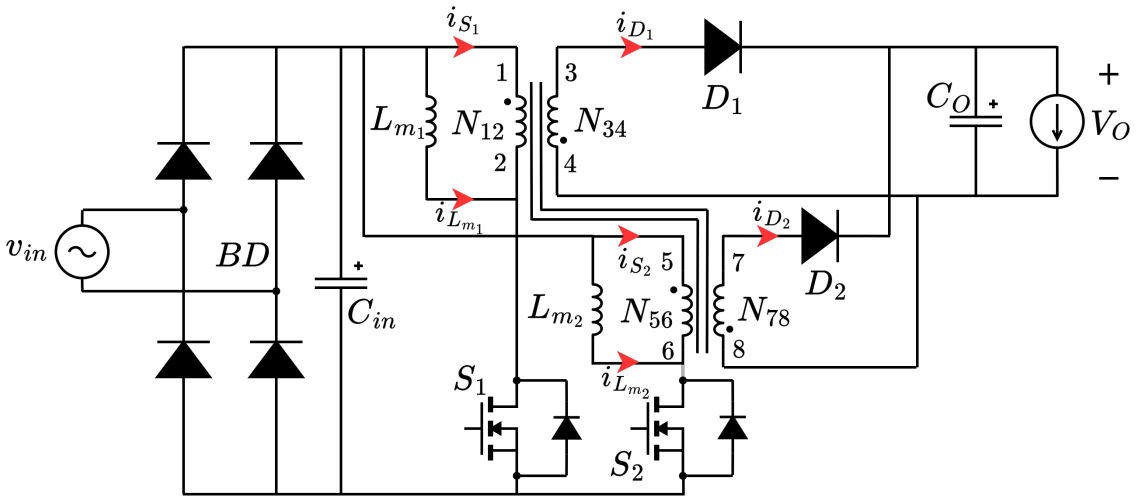


Figure 3.1: The proposed two-phase interleaved flyback PFC rectifier.

The proposed integrated transformer was designed for a two-phase interleaved flyback PFC rectifier operating in discontinuous conduction mode (DCM). Each switch S_1, S_2 in Figure 3.1 operates uniformly at a constant duty cycle with 180-degree phase shift.

To enable independent operation per phase without magnetic interference, any coupling between the two transformers is eliminated. In other words, the transformers must not influence each other in each phase when a winding is wound around a leg and current is passed through it [21], [22]. The following winding structure makes it possible for the conventional, two-separate core structure to be integrated into one, single core, allowing for downsizing of magnetic components. Details of the influence of the winding structure on the operating principle of this transformer will be explained further.

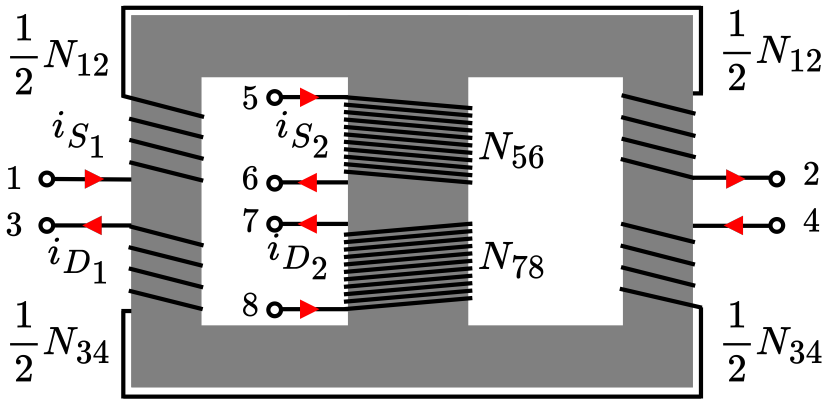


Figure 3.2: The proposed non-coupled integrated transformer in this study.

The winding structure is shown in 3.2. Half of the total number of turns for transformer N_{12} and N_{34} are wound at each outer leg with opposite polarity. Moreover, N_{56} and N_{78} are also wounded with opposite polarity at the center leg. As an additional note, the magnetic resistance of each outer leg is also made equal.

The behavior of various electrical quantities and also the flux density will be examined from now on. For the flux density in the core, especially that the transformer can operate separately in two phases, the principle of superposition can be used to examine the flux behavior in each leg of the transformer for each phase. This is important in deciding the saturation magnetic flux density, which is important for the design of the integrated transformer. From the transformer in Figure 3.3, B_l, B_r, and B_c are the magnetic flux densities in the left, right, and center legs respectively. The transformers can then be separated into each phase as per the principle of superposition. As shown in Figure 3.4a and 3.4b, B'_l, B'_r, and B'_c represent the flux densities in the positive AC Cycle, while B''_l, B''_r, and B''_c are the flux densities in the negative AC Cycle.

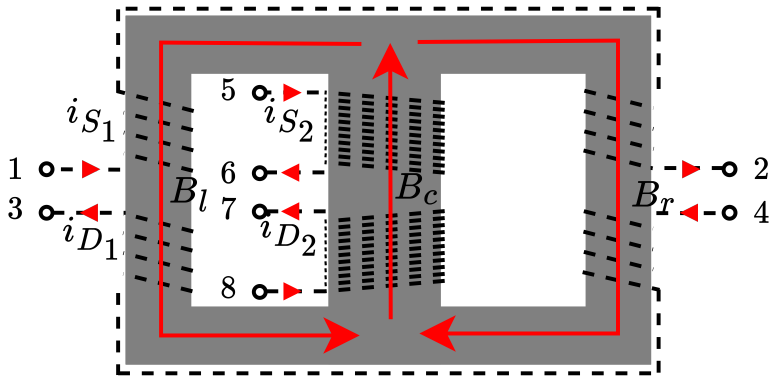
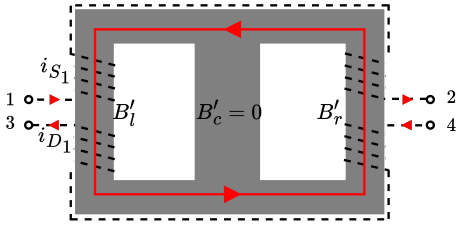
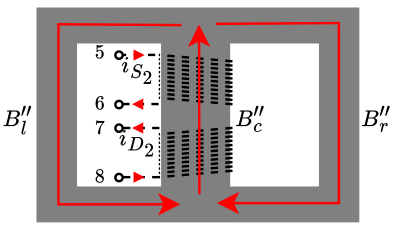


Figure 3.3: Flux densities inside the proposed integrated transformer. B_l represents the flux density in the left leg, B_r represents the flux density in the right leg, and B_c represents the flux density in the right leg.



(a) Outer transformer $N_{12}N_{34}$ with flux densities.



(b) Center leg transformer $N_{56}N_{78}$ with flux densities .

Figure 3.4: Integrated transformer separated according to AC operation cycle.

By the principle of superposition, shown in Equation 3.1 are the corresponding expressions for the superimposed flux densities in each of the legs.

$$\begin{cases} B_l &= B'_l + B''_l \\ B_r &= B'_r - B''_r \\ B_c &= B''_c \end{cases} \tag{3.1}$$

The following assumptions were made in this analysis:

1. The cross-sectional area of the central leg in the EE core is twice that of an outer leg, as per commercial standard, causing $N_{12} = 2N_{56}$ and $N_{34} = 2N_{78}$.

2. The magnetizing current and magnetic flux density are proportional, with B'_{max} as the maximum flux density in each transformer at each phase at the maximum magnetizing inductance current.

3. The saturation magnetic flux B'_{max} in each separate transformer during each phase is expected to be consistent. This is because the magnetizing inductance of each transformer (L_{m_1}, L_{m_2} in Figure 3.1) is set the same for each phase as well, and each phase is driven at the same conditions.

4. To simplify the analysis, the resonant operation between the magnetizing inductance and the parasitic capacitance of the switch is not considered.

3.1 Circuit and Transformer Operational Analysis

Crucial in the operational analysis of the circuit is the discharge time of the transformer after the switches are turned off in the primary side. The duty cycle D influences how fast the transformer discharges, and will be explored in this section.

The circuit and transformer is first analyzed according to the condition where $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$, where n represents the turns ratio, V_O represents the

output voltage, and $V_{\text{in, max}}$ represents the amplitude of the input voltage. In other words, one transformer discharges before the other fully charges, or its magnetizing inductance current reaches the peak.

A compilation of the essential waveforms involved in the circuit and transformer operation is shown in Figure 3.5. The quantities v_{gs_1} and v_{gs_2} represent the corresponding gate-source voltages, i_{S_1} and i_{S_2} represent the corresponding switch currents, i_{D_1} and i_{D_2} represent the corresponding secondary side diode currents, $i_{L_{m_1}}$ and $i_{L_{m_2}}$ represent the corresponding magnetizing inductance currents. The quantities B_l, B_r, and B_c are the magnetic flux densities in the left, right, and center legs respectively. Moreover, B_l', B_r', and B_c' represent the flux densities in the positive AC Cycle, while B_l'', B_r'', and B_c'' are the flux densities in the negative AC Cycle.

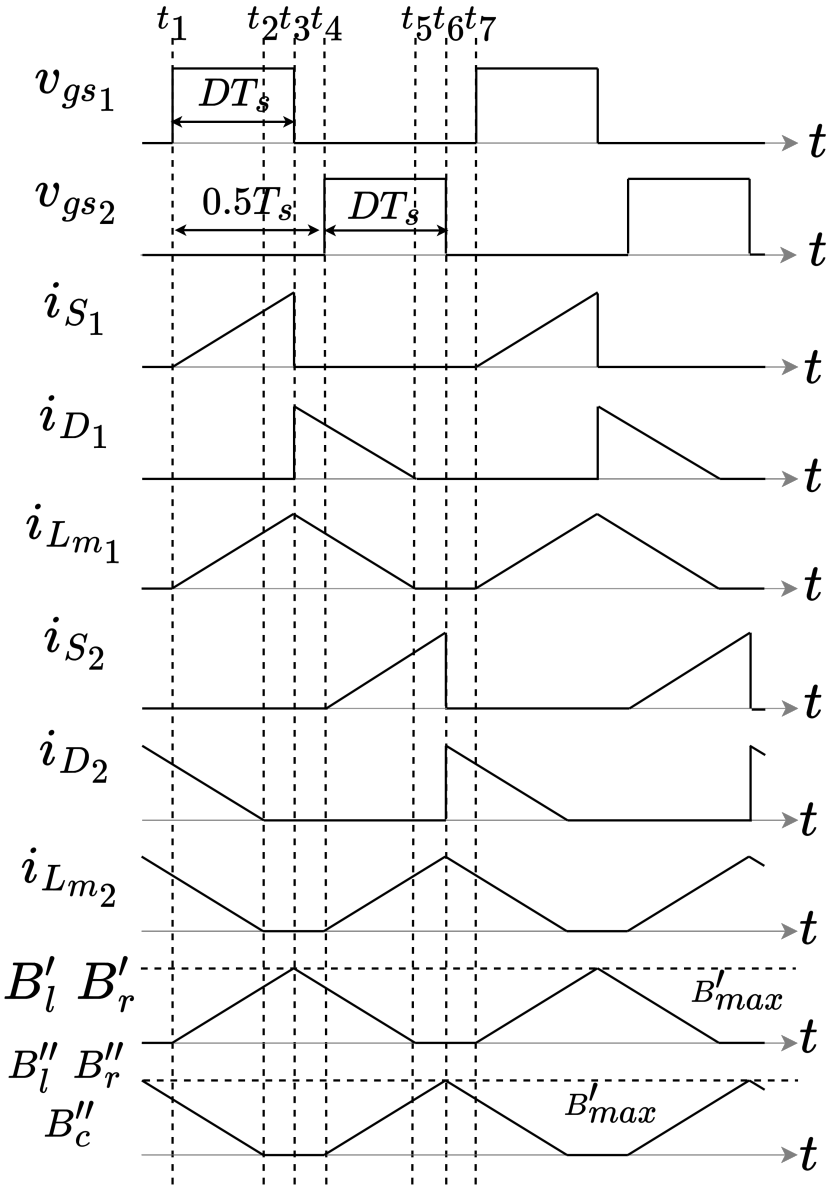


Figure 3.5: Key electrical and flux density waveforms of the circuit when $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$.

The circuit operates in the following manner in the positive AC cycle. The key waveforms are shown in Figure 3.5.

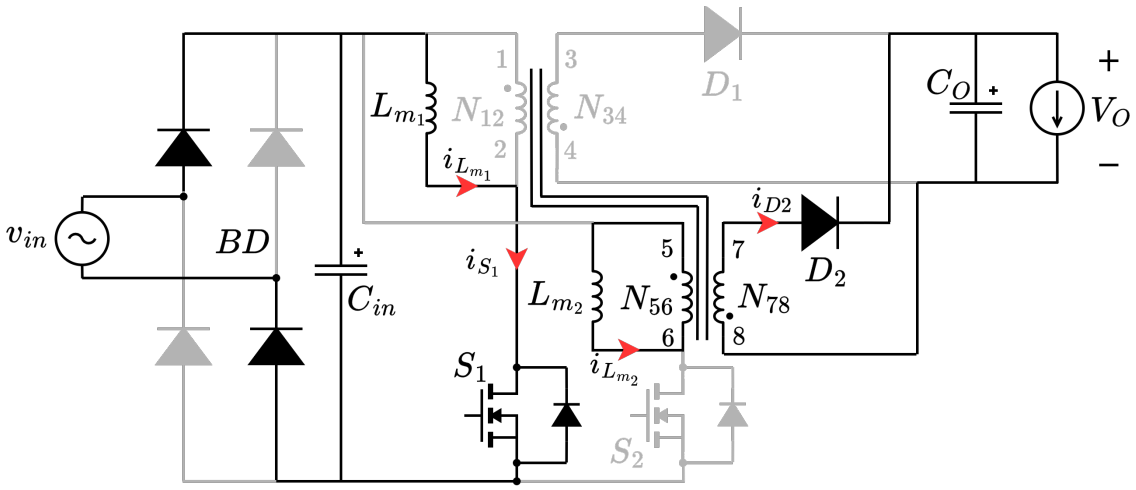


Figure 3.6: Operation of the proposed circuit with the transformer at Mode 1, $t_1 < t < t_2$.

Mode 1, $t_1 < t < t_2$: With S_1 on and S_2 off, the following happens. The transformer $N_{56}N_{78}$ continues to discharge until its corresponding magnetizing inductance current $i_{L_{m_2}}$ and secondary side current i_{D_2} reach zero. The flux densities B_l'', B_r'', and B_c'' decrease linearly as well until they reach zero. The current i_{S_2} is zero since the corresponding switch S_2 is off. On the other hand, since transformer $N_{12}N_{34}$ is storing energy in its magnetizing inductance, then its current $i_{L_{m_1}}$ should be linearly increasing, and also the corresponding flux densities B_l' and B_r'. Current i_{S_1} must be increasing linearly as well. Moreover, in this phase, since $N_{12}N_{34}$ is storing energy, then i_{D_1} must be zero. A representation of the active components in this mode are shown in Figure 3.6.

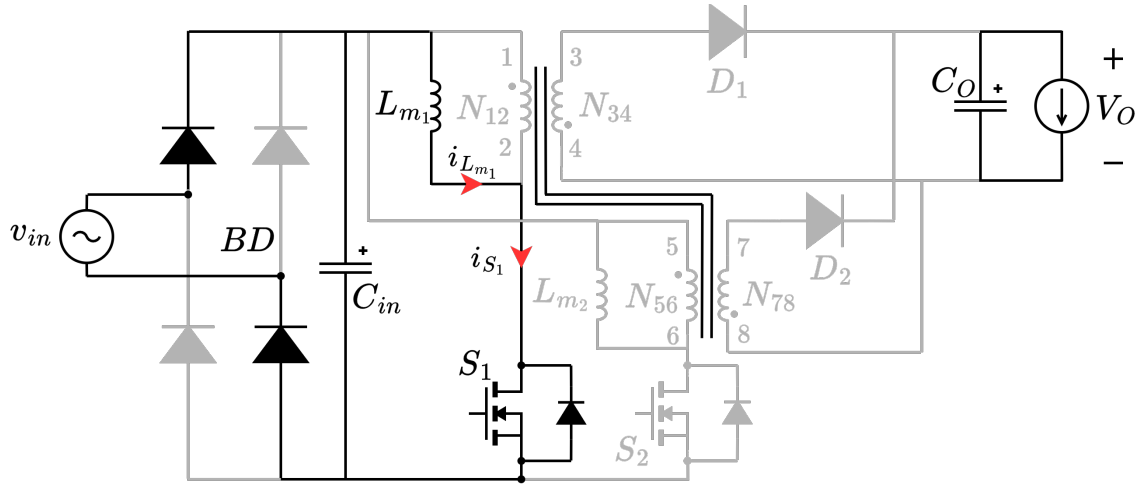


Figure 3.7: Operation of the proposed circuit with the transformer at Mode 2, $t_2 < t < t_3$.

Mode 2, $t_2 < t < t_3$: Flux densities B_l' and B_r', currents i_{S_1}, and $i_{L_{m_1}}$ reach their peak, as the magnetizing inductance L_{m_1} becomes fully charged, ready for energy transfer. Here, the quantities related to $N_{56}N_{78}$, which are the currents i_{D_2}, $i_{L_{m_2}}$, and the flux densities B_l'', B_c'' and B_r'', remain at zero, since it is inactive during this time period. A representation of the active components in this mode are shown in Figure 3.7.

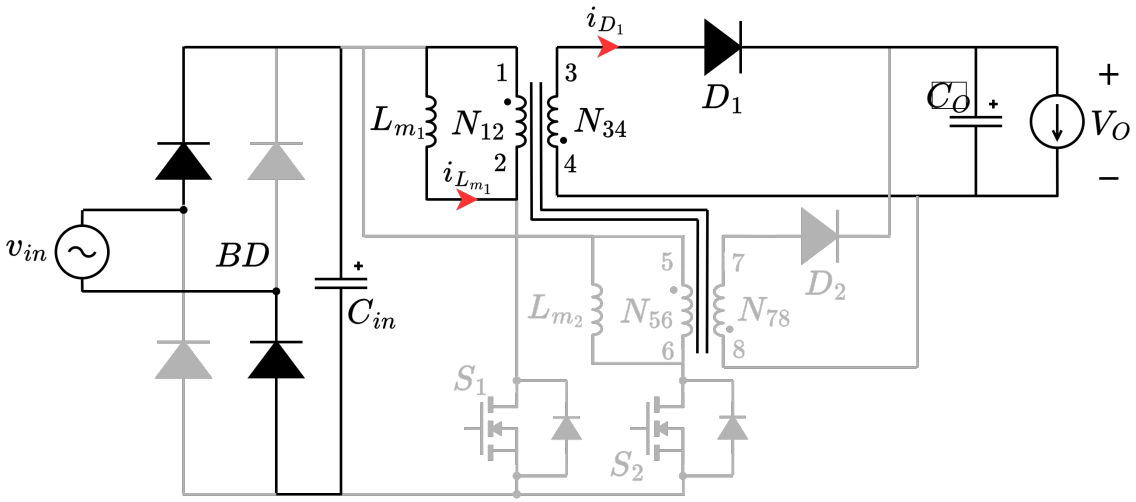


Figure 3.8: Operation of the proposed circuit with the transformer at Mode 3, $t_3 < t < t_4$.

Mode 3, $t_3 < t < t_4$: Switch S_1 turns off, signifying that $i_{L_{m_1}}$ has reached its peak value. The transformer $N_{12}N_{34}$ has finished charging and has started to discharge. This explains why currents i_{D_1}, $i_{L_{m_1}}$, and flux densities B_l', and B_r' start decreasing linearly. The corresponding currents and flux densities to $N_{56}N_{78}$ remain at zero, since it is inactive during this time period and S_2 is still off. A representation of the active components in this mode are shown in Figure 3.8.

The circuit operates in the following manner in the negative AC cycle. The key waveforms are shown in Figure 3.5.

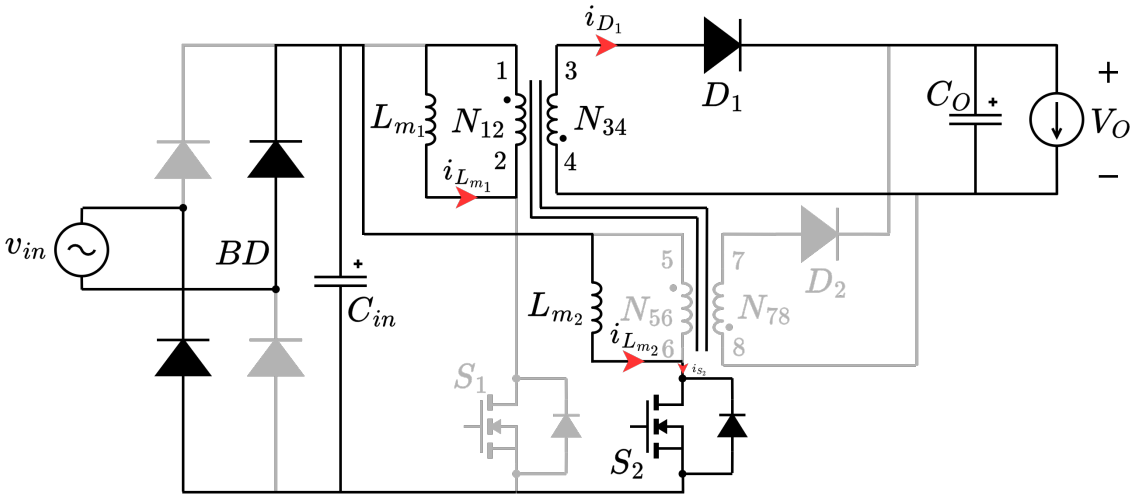


Figure 3.9: Operation of the proposed circuit with the transformer at Mode 4, $t_4 < t < t_5$.

Mode 4, $t_4 < t < t_5$: Contrary to Mode 1, S_2 is on and S_1 is off. The transformer $N_{12}N_{34}$ continues to discharge until currents $i_{L_{m_1}}$ and i_{D_1} reach zero. The flux densities B_l', and B_r' decrease linearly as well until they reach zero. Current i_{S_1} is zero since S_1 is off. On the other hand, since transformer $N_{56}N_{78}$ is storing energy in its magnetizing inductance, then $i_{L_{m_2}}$ should be linearly increasing, as well as B_c'', B_l'' and B_r''. Current i_{S_2} must be linearly increasing as well. Moreover, in this phase, since $N_{56}N_{78}$ is storing energy, then current i_{D_2} must be zero. A representation of the active components in this mode are shown in Figure 3.9.

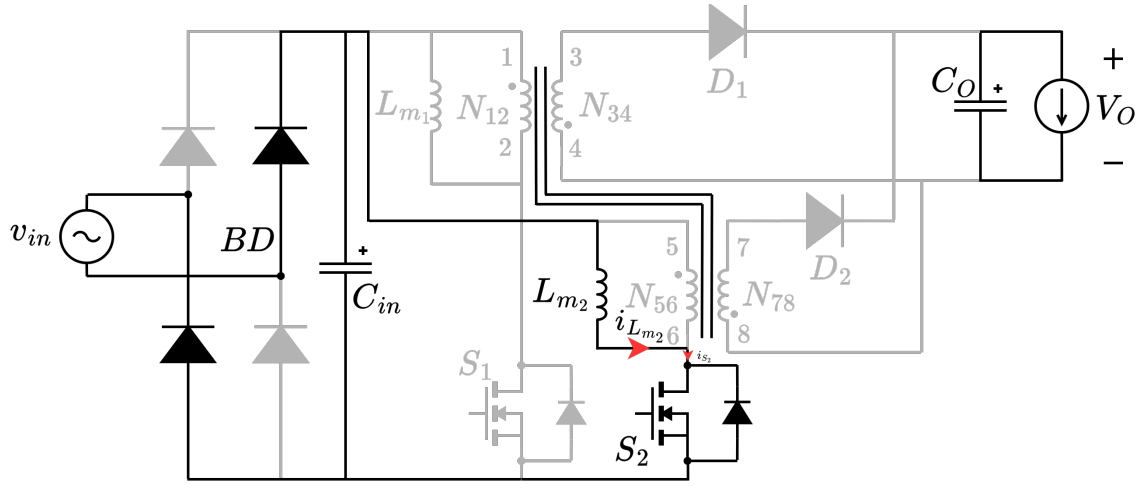


Figure 3.10: Operation of the proposed circuit with the transformer at Mode 5, $t_5 < t < t_6$.

Mode 5, $t_5 < t < t_6$: Flux densities B_l'', B_c'' and B_r'', currents i_{S_2}, and $i_{L_{m_2}}$ reach their peak, as the magnetizing inductance L_{m_2} becomes fully charged, ready for energy transfer. Here, the quantities related to $N_{12}N_{34}$ remain at zero, which are the currents i_{D_1}, $i_{L_{m_1}}$, and the flux densities B_l' and B_r'', since it is inactive during this time period. A representation of the active components in this mode are shown in Figure 3.10.

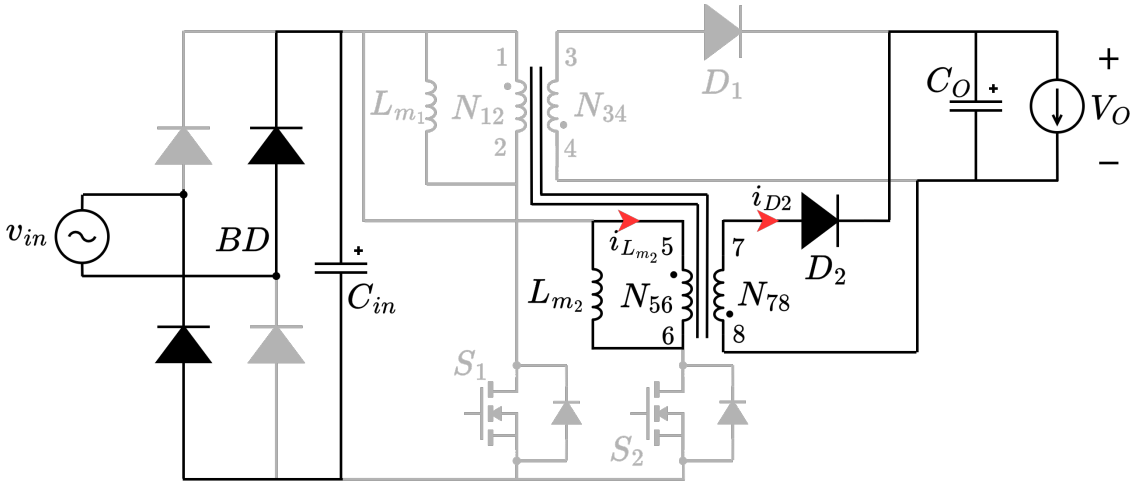


Figure 3.11: Operation of the proposed circuit with the transformer at Mode 6, $t_6 < t < t_7$.

Mode 6, $t_6 < t < t_7$: Switch S_2 turns off, signifying that current $i_{L_{m_2}}$ has reached its peak value. The transformer $N_{56}N_{78}$ has finished charging and has started to discharge. Currents i_{D_2}, $i_{L_{m_2}}$, and the flux densities B_l'', B_c'' and B_r'' start decreasing linearly. The corresponding currents and flux densities for transformer $N_{12}N_{34}$ remain at zero, since it is inactive during this time period and switch S_1 is still off. A representation of the active components in this mode are shown in Figure 3.11.

The peak value of $i_{S,\text{pk}}$ in a cycle as shown in Figure 3.5 is obtained using Equation 3.2.

$$i_{S,\text{pk}} = \frac{v_{in} \cdot DT_s}{L_m} \tag{3.2}$$

where v_{in} is the input voltage, D is the duty cycle, T_s is the switching period, and L_m is the magnetizing inductance of the transformer involved.

The peak flux B_{pk} in a cycle as shown in Figure 3.5 is obtained using Equation 3.3.

$$B_{\text{pk}} = \frac{L_m i_{S,\text{pk}}}{N_p A} \tag{3.3}$$

where N_p is the number of turns in the primary side of the active transformer and A is the cross-sectional area of the leg of the transformer involved.

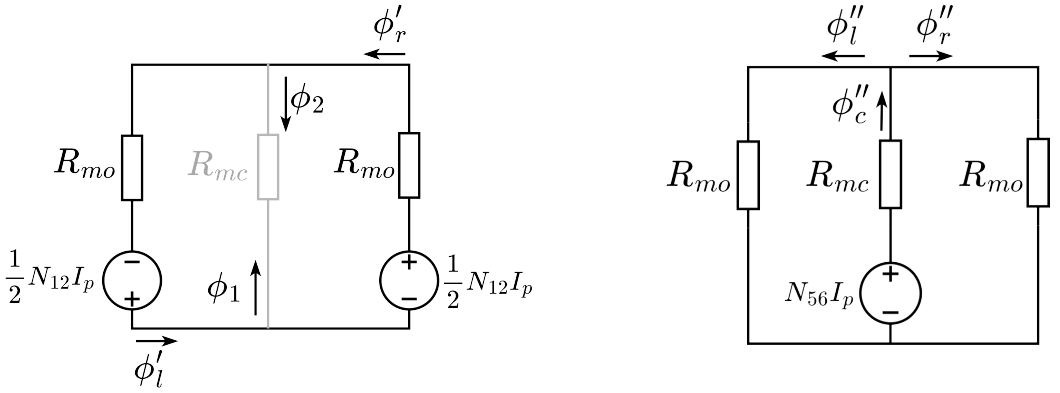
No coupling implicates that the operation of one transformer does not affect the other during the discharge stage. More specifically, the operation of one transformer does not affect the flux density of its partner transformer. This then implies the following:

1. The i_{L_m}, B waveforms, which are directly related to i_D should be linear in increase and decrease.

2. After discharge, the mentioned waveforms should stay at 0.

3. The inactive transformer should have $i_D = 0$.

Therefore, one could examine i_{D_1} and i_{D_2} explicitly to determine if the circuit operation includes coupling. This will be used in the simulation and results verification section. This phenomenon is easily seen in modes 3 and 6.



(a) Outer transformer $N_{12}N_{34}$. **(b)** Center transformer $N_{56}N_{78}$.

Figure 3.12: Equivalent magnetic circuits for the proposed transformer.

The influence of the proposed winding to the operation of the transformer at modes 3 and 6 are further examined.

For Mode 3, the transformer being realized is shown in Figure 3.4a and the corresponding magnetic circuit is shown also in Figure 3.12a. For the positive cycle, since the magnetomotive force and magnetic resistance of both outer legs are equal, the magnetic flux flowing to the center leg is canceled, resulting in no electromotive force being generated in transformer $N_{56}N_{78}$ [21], [22]. Therefore, the transformer

formed by the outer legs can operate independently from the transformer in the center leg.

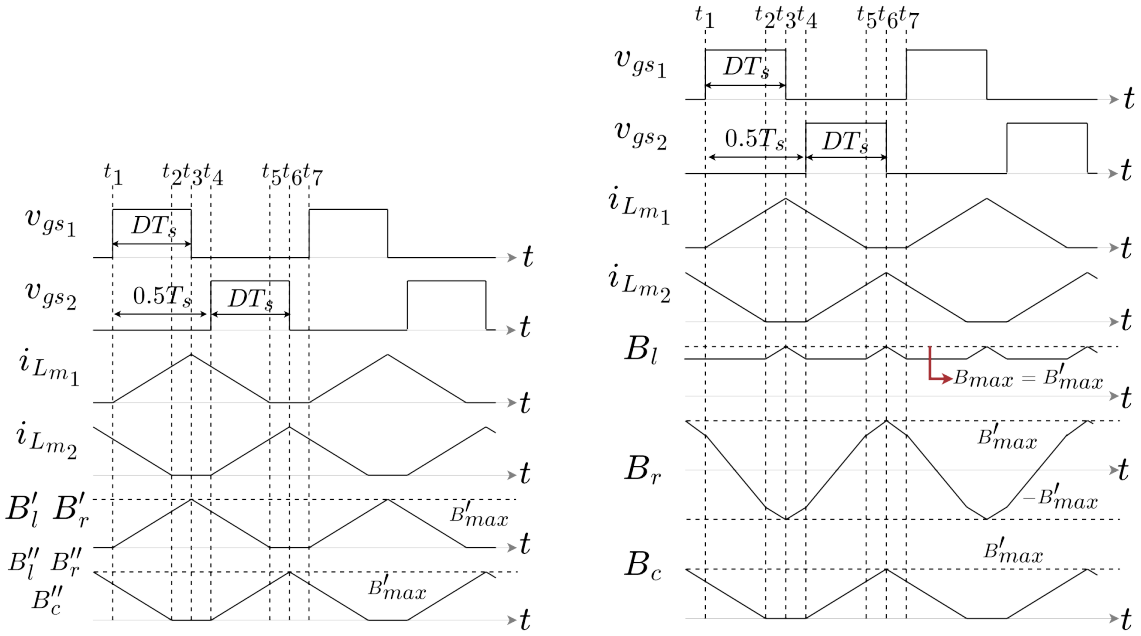
For Mode 6, the transformer being realized is shown in Figure 3.4b and the corresponding magnetic circuit is shown also in Figure 3.12b. For the negative cycle, since the magnetic resistance of both outer legs is equal, the magnetic flux is divided equally into both outer legs. Furthermore, since the number of turns of both outer legs is equal and wound in an opposite polarity, the electromotive force generated in transformer $N_{12}N_{34}$ is canceled [21], [22]. Therefore, the transformer formed by the center leg can operate independently from the transformer in the outer leg.

The proposed winding configuration results in a zero-coupling coefficient between the transformers, allowing them to operate separately in two phases. Therefore, the two transformers can be integrated into one core, downsizing the magnetic components in the circuit.

3.2 Saturation Magnetic Flux Density Analysis

This section will detail the total magnetic flux density in the core in different operation patterns, as affected by the duty cycle. As mentioned in the previous section, the principle of superposition can be used to obtain the total, shown in Equation 3.1.

The other two operational patterns will not be explained in detail, yet the superimposed flux density waveforms are shown in this section, along with the saturation magnetic flux densities of each case, which is important for the transformer design.



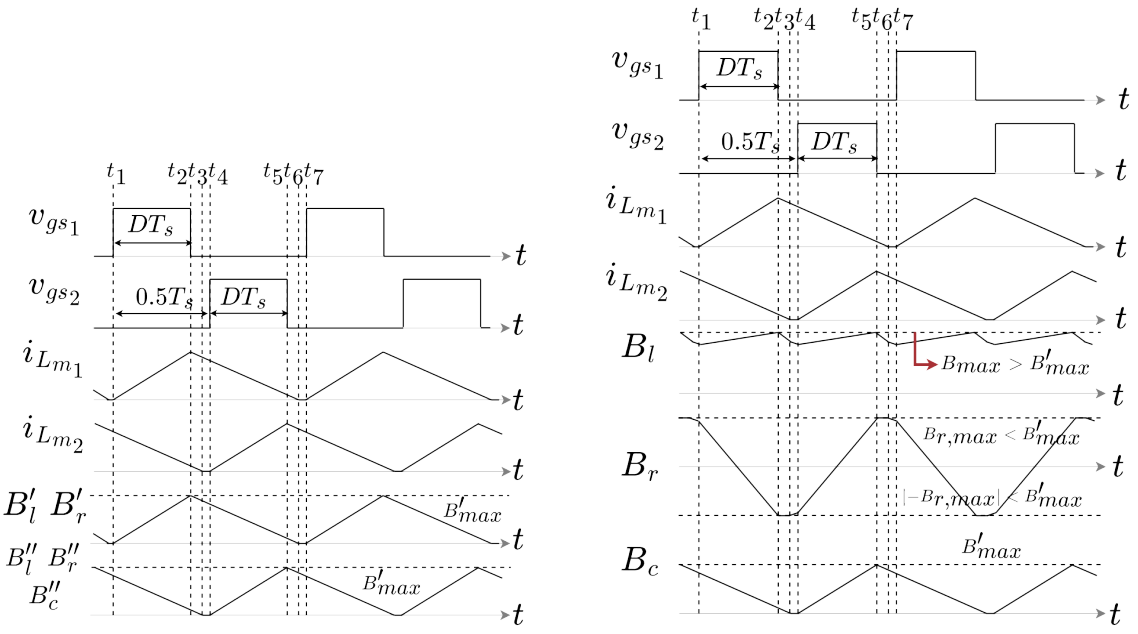
(a) Separated waveforms. **(b)** Superimposed waveforms.

Figure 3.13: Magnetic flux density waveforms for Case 1: $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$.

For Case 1: $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$. Here, one transformer discharges before the other fully charges, or its magnetizing inductance reaches the peak.

Superimposing the flux densities in Figure 3.13a with each other will then result in the following waveforms in Figure 3.13b. The left and right outer legs are not capable of magnetic flux cancellation. However, the peak fluxes can be seen in all legs. Therefore, the the saturation magnetic flux density of the core across all the phases is shown in Equation 3.4.

$$B_{max} = B'_{max} \tag{3.4}$$



(a) Separated waveforms. **(b)** Superimposed waveforms.

Figure 3.14: Magnetic flux density waveforms for Case 2: $D \leq 0.5$ and $D > 0.5\frac{nV_O}{V_{\text{in, max}}}$.

For Case 2: $D \leq 0.5$ and $D > 0.5\frac{nV_O}{V_{\text{in, max}}}$. Here, one transformer does not discharge fully before the other fully charges, or its magnetizing current reaches the peak.

Superimposing the flux densities in Figure 3.14a with each other will then result in the following waveforms in Figure 3.14b. The left and right outer legs are not capable of magnetic flux cancellation. However, the peak flux density can only be seen in the left outer leg, at t_2 and t_5. The saturation magnetic flux density across all phases can be derived as follows and is shown in Equation 3.10.

At t_2, B'_l is B'_{max}.

At t_2, B''_l is also the B'_l at $t = t_5$:

$$B''_l(t_2) = B'_l(t_5) = B'_{max} - \frac{B'_{max}}{t_6 - t_2}(t_5 - t_2) \tag{3.5}$$

From Figure 3.14 , $t_5 = 0.5T_s + t_2 = 0.5T_s + DT_s$ and $t_2 = DT_s$.

To get t_6, one can examine the increase and decrease of the magnetizing inductance current during charge and discharge of the transformer. The peak, $i_{L_{m_1},\,\text{pk}}$,

can be rewritten in terms of the charging and discharging side:

$$I_{L_{m_1},\,\mathrm{pk}} = \frac{V_{\mathrm{in,\,max}}}{L_m} DT_s = \frac{nV_O}{L_m}(t_6 - t_2) \tag{3.6}$$

Solving equation 3.6 for t_6 yields:

$$t_6 = \frac{V_{\mathrm{in,\,max}}}{nV_O} DT_s + t_2 \tag{3.7}$$

$$= \frac{V_{\mathrm{in,\,max}}}{nV_O} DT_s + DT_s \tag{3.8}$$

Using Equation 3.5 and 3.8, the B_l'' at $t = t_2$ is given as:

$$B_l''(t_2) = B_l'(t_5) = B'_{max} - \frac{0.5 B'_{max} nV_O}{V_{\mathrm{in,\,max}} D} \tag{3.9}$$

And therefore, the superimposed B_{max} found at the left outer leg is given by Equation 3.10, at t_2 and t_5:

$$B_{max} = B''_{l_{max}} = \left(2 - \frac{0.5 nV_O}{V_{\mathrm{in,\,max}} D}\right) B'_{max} \tag{3.10}$$

To realize the boundary condition between case 1 and 2, as mentioned, the active transformer must discharge before the other transformer finsihes charging by reaching its peak magnetizing inductance current.

The discharge time $t_{\mathrm{discharge}}$ of the transformer according to Faraday's Law will be given by Equation 3.11.

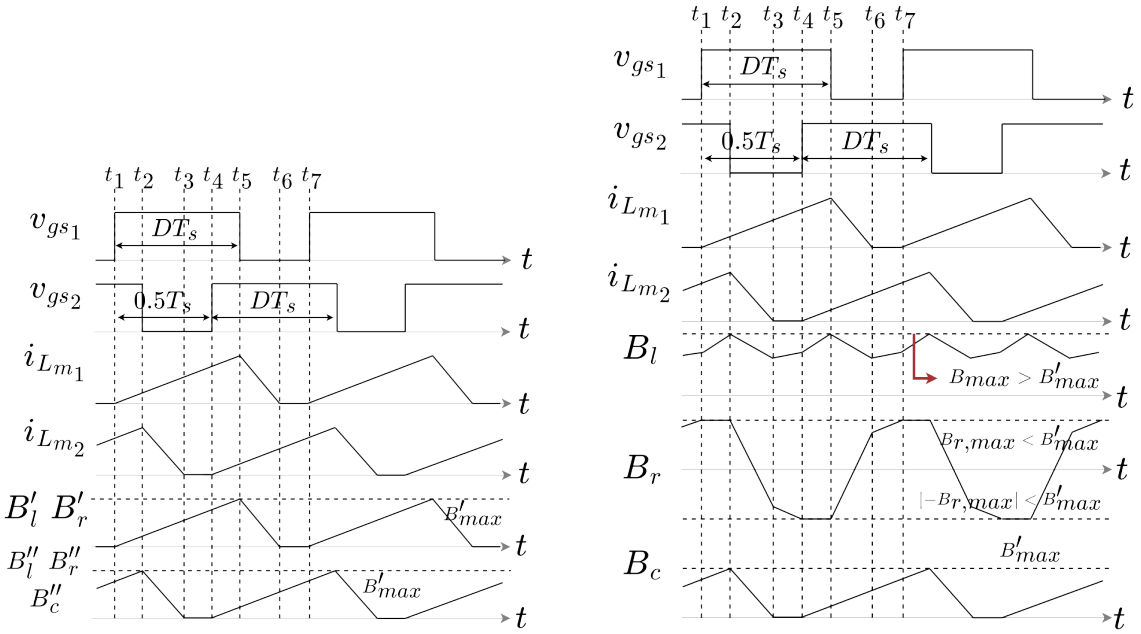
$$t_{\mathrm{discharge}} = \frac{V_{\mathrm{in,\,max}}}{nV_O} DT_s \tag{3.11}$$

After setting $t_{\mathrm{discharge}} \leq 0.5T_s$, along with $D < 0.5$, the boundary condition between

Case 1 and 2 is given by Equation 3.12.

$$D \leq 0.5 \frac{nV_O}{V_{\text{in, max}}} \tag{3.12}$$

where n is the turns ratio, V_O is the output voltage, and $V_{\text{in, max}}$ is the amplitude of the input voltage.



(a) Separated Waveforms. **(b)** Superimposed Waveforms

Figure 3.15: Magnetic flux density waveforms for Case 3: $D > 0.5$.

For Case 3: $D > 0.5$. Here, the transformers charge in succession and discharge before the next charge cycle of the other transformer.

Similar to Pattern 2, superimposing the fluxes in Figure 3.15a with each other will then result in the following waveforms in Figure 3.15b. The left and right outer legs are not capable of magnetic flux cancellation.The left and right outer legs are not capable of magnetic flux cancellation. However, the peak flux density can only be seen in the left outer leg, at t_2 and t_5. The saturation magnetic flux density across all phases can be derived as follows and is shown in Equation 3.15.

At t_2, B'_l is also the B''_l at $t = t_5$:

$$B'_l(t_2) = B''_l(t_5) = \frac{B'_{max}}{DT_s}(t_5 - t_4) \tag{3.13}$$

At t_2, B''_l is B'_{max}. From Figure 3.15, $t_5 = DT_s$ and $t_4 = 0.5T_s$.

Therefore, $B'_l(t_2)$ is as follows:

$$B'_l(t_2) = B''_l(t_5) = \frac{B'_{max}}{DT_s}(DT_s - 0.5T_s) \tag{3.14}$$

And therefore, the superimposed B_{max} found at the left outer leg is given by, at t_2 and t_5:

$$B_{max} = B''_{l_{max}} = \left(1 + \frac{D - 0.5}{D}\right) B'_{max} \tag{3.15}$$

3.3 Miniaturization Effect of the Proposed Transformer

This section investigates the potential contribution of the proposed transformer to the volume reduction of the magnetic component. This section utilizes a part of the analysis method by [23]. This section will then detail too the volume and area products of the conventional and proposed transformer.

Firstly, the area-product of the conventional and proposed transformer is derived. The Area-Product (AP) Method is a core selection method wherein the transformer geometric dimensions are decided based on the circuit specifications by a relation between relevant quantities, such as current density and magnetizing inductance [24], making it a suitable way to determine the volume of the transformer. More specifically, the product of the window area A_w and magnetic path cross sectional area A_c, called the Area Product (AP) can be related to the circuit parameters. Choosing a core that has a bigger AP than the AP decided by the circuit parameters is ideal.

The analysis can be made simpler by making A_w and A_c a square shape, which is shown here in Figure 3.16 [23]. An EE-Core was chosen for this study, as it is one of the most readily available magnetic cores in the market. For the EE-core transformer, it is assumed that the area of the center leg is twice of the outer one, corresponding to the industry standard.

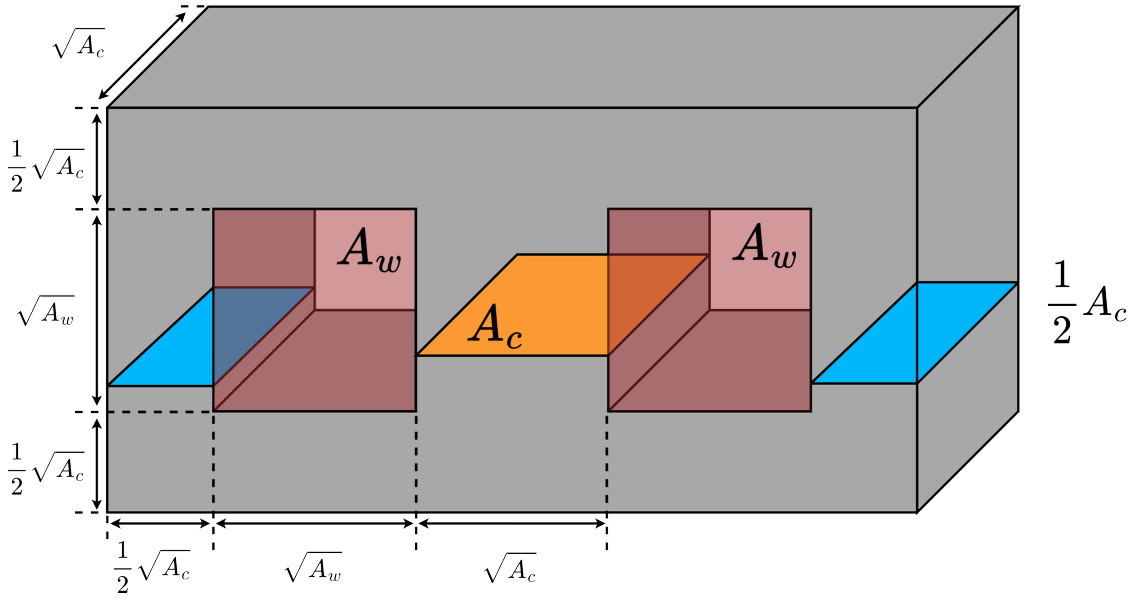


Figure 3.16: EE-core with a simplified A_w and A_c [23].

For the following analyses, the variables used are shown in Table 3.1.

Table 3.1: Parameters used in the volume analysis.

Variable	Meaning
L_m	Magnetizing Inductance
$I_{\text{p, pk}}$	Primary Side Peak Current
$I_{\text{p, rms}}$	Primary Side RMS Current
$I_{\text{s, rms}}$	Secondary Side RMS Current
n	Turns Ratio
J_m	Maximum Current Density
B_{max}	Maximum Flux Density
K_m	Occupancy Ratio

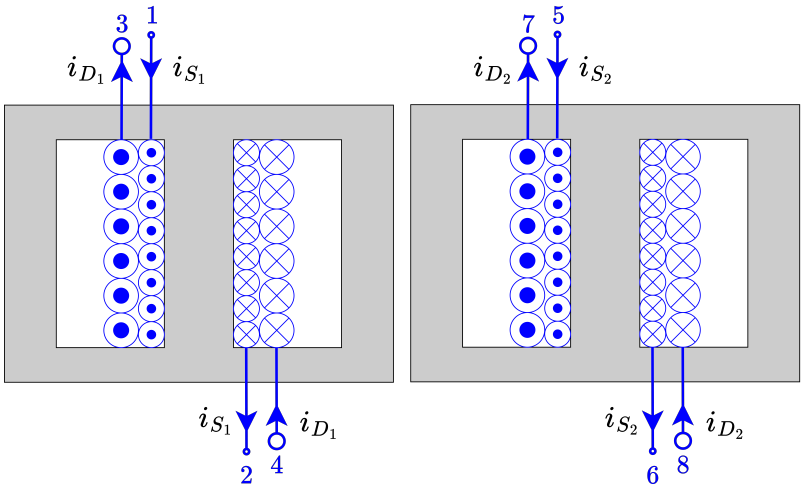


Figure 3.17: Representation of the conventional transformer with windings.

Conventional Transformer. An analysis for the conventional transformer was first performed. Here, $N_{12} = N_{56}$ and $N_{34} = N_{78}$. The cross-section of the conventional transformer is shown in Figure 3.17. One of the two transformers is analyzed first. Using Faraday's Law, an expression for A_c can be obtained as shown in Equation 3.17:

$$N_{12} B_{max} A_c = L_m I_{\text{p, pk}} \tag{3.16}$$

$$A_c = \frac{L_m I_{\text{p, pk}}}{N_{12} B_{max}} \tag{3.17}$$

On the other hand, the window area can be expressed in terms of the cross-sectional area of the windings as shown in Equation 3.20, with Faraday's Law, by taking it as a fraction of the window area A_w, denoted by the occupation factor K_u:

$$A_w K_u = N_{12} \frac{I_{\text{p, rms}}}{J_m} + N_{56} \frac{I_{\text{s, rms}}}{J_m} \tag{3.18}$$

$$A_w K_u = N_{12} \frac{I_{\text{p, rms}}}{J_m} + \frac{N_{12}}{n} \frac{I_{\text{s, rms}}}{J_m} \tag{3.19}$$

$$A_w = \frac{N_{12}}{J_m K_u} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right) \tag{3.20}$$

And therefore, the area product of one conventional transformer using Equation 3.17 and 3.20 is:

$$AP = A_c A_w = \frac{L_m I_{\text{p, pk}} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right)}{B_m J_m K_u} \tag{3.21}$$

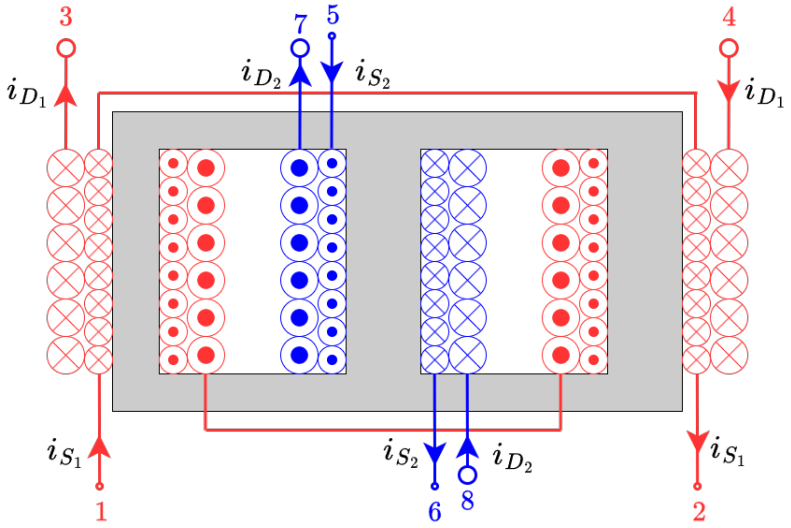


Figure 3.18: Representation of the proposed transformer with windings.

Proposed Integrated Transformer. An analysis for the integrated transformer was performed next. The cross-section of the integrated transformer is shown in Figure 3.18. Using Faraday's Law, an expression for A_c can be obtained in two ways as shown in Equation 3.24.

$$N_{12} B_{max} \frac{A_c}{2} = L_m I_{\text{p, pk}} \tag{3.22}$$

$$N_{56} B_{max} A_c = L_m I_{\text{p, pk}} \tag{3.23}$$

$$A_c = \frac{2 L_m I_{\text{p, pk}}}{N_{12} B_{max}} = \frac{L_m I_{\text{p, pk}}}{N_{56} B_{max}} \tag{3.24}$$

On the other hand, similarly, the cross-sectional area of the windings can be obtained using the following expression, with Faraday's Law. It is important to note that because of the assumption that the middle leg cross-sectional area is twice

than that of the outer leg, then $N_{12} = 2N_{56}$ and $N_{34} = 2N_{78}$. Moreover, for the proposed transformer, both half of the primary side winding and also the secondary side winding are occupying the window at the same time. This leads to the following expression in Equation 3.27:

$$A_w K_u = \left(N_{56} \frac{I_{\text{p, rms}}}{J_m} + N_{78} \frac{I_{\text{s, rms}}}{J_m} \right) + \frac{1}{2} \left(N_{12} \frac{I_{\text{p, rms}}}{J_m} + N_{34} \frac{I_{\text{s, rms}}}{J_m} \right) \tag{3.25}$$

$$A_w K_u = N_{12} \frac{I_{\text{p, rms}}}{J_m} + \frac{N_{12}}{n} \frac{I_{\text{s, rms}}}{J_m} \tag{3.26}$$

$$A_w = \frac{2N_{56}}{J_m K_u} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right) \tag{3.27}$$

And therefore, the area product of the integrated transformer is:

$$AP = A_c A_w = \frac{\mathbf{2} L_m I_{\text{p, pk}} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right)}{B_m J_m K_u} \tag{3.28}$$

From Equation 3.21 and 3.28, the area product of the integrated transformer is twice that of the conventional transformer.

$$AP_{integ.} = 2AP_{conv.} \tag{3.29}$$

The volume of the EE-core illustrated in Figure 3.16 can be written in terms of the area product AP and $x = \frac{A_c}{A_w}$:

$$\text{Volume} = 2AP^{\frac{3}{4}} \left(x^{\frac{3}{4}} + 2x^{\frac{1}{4}} + x^{-\frac{1}{4}} \right) \tag{3.30}$$

Finally, theoretically, the ratio of the volume of the integrated transformer and the proposed transformer can be calculated using Equations 3.21, 3.28, 3.29, and 3.30 as shown in Equation 3.31. It is important to note that the conventional transformer is composed of two cores of the same kind, and thus its volume is twice

than that of one core.

$$\frac{\text{Volume}_{integ.}}{\text{Volume}_{conv.}} = \frac{\mathbf{2} \cdot 2AP_{conv.}^{\frac{3}{4}} \left(x^{\frac{3}{4}} + 2x^{\frac{1}{4}} + x^{-\frac{1}{4}} \right)}{2AP_{integ.}^{\frac{3}{4}} \left(x^{\frac{3}{4}} + 2x^{\frac{1}{4}} + x^{-\frac{1}{4}} \right)} = 0.84 \tag{3.31}$$

This then shows that the proposed integrated transformer provides a 16% volume reduction compared to the conventional transformer.

Moreover, the expressions for $I_{\text{p, rms}}$ and $I_{\text{s, rms}}$ are shown in Equations 3.32 and 3.33. These can be used, as the circuit used in this study works similarly to that of the circuit in [11].

$$I_{\text{p, rms}} = \frac{V_{\text{in, max}} D}{L_m f_s} \sqrt{\frac{D^3}{6}} \tag{3.32}$$

$$I_{\text{s, rms}} = \frac{V_{\text{in, max}} D}{L_m f_s} \sqrt{\frac{5nD^3}{9\pi} \cdot \frac{V_{\text{in, max}}}{V_O}} \tag{3.33}$$

Chapter 4

Transformer Design Method

This section provides a design method for the proposed integrated transformer. A core is selected using the area product method as described in Chapter 3. The method for choosing a core will not be covered anymore in this chapter. Moreover, it is assumed that the magnetizing inductances for the two transformers are equal $L_{m1} = L_{m2} = L_m$.

4.1 Turns Ratio

Firstly, a turns ratio n between the primary and secondary side is first decided. To make it simple, a turns ratio $n = \frac{N_P}{N_S}$ is decided based on the maximum voltage specifications of the chosen switching device. Fundamentally, the switch should withstand the peak voltage due to the effect of noise, magnetic induction, and the input voltage itself too. The following value of n can be obtained from the inequality shown in Equation 4.2.

$$V_{\text{DS, max}} \geq nV_O + V_{\text{in, max}} + V_{\text{spike}} + V_{margin} \tag{4.1}$$

$$n \leq \frac{V_{\text{DS, max}} - V_{\text{in, max}} - V_{\text{spike}} - V_{margin}}{V_O} \tag{4.2}$$

Here, $V_{\text{DS, max}}$ represents the maximum drain-source voltage specification of the switching device, V_O represents the output voltage, $V_{\text{in, max}}$ represents the peak of the input voltage, V_{spike} and V_{margin} represent some leeway for surge voltage and noise that the device may withstand and are typically set to 100 V in various design applications.

4.2 Saturation Magnetic Flux Density

As have been discussed in Chapter 3, a specific magnetic flux pattern occurs in the transformer depending on the duty cycle of the switches. The desired flux operating pattern is first decided then afterwards a duty cycle for the circuit can be decided. There are therefore, three cases for the saturation magnetic flux density B'_{max} that occurs in the transformer in each phase, as shown in Equations 4.3 to 4.5.

Case 1: $D \leq 0.5$ with $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$

$$B'_{max} = B_{max} \tag{4.3}$$

Case 2: $D \leq 0.5$ with $D > 0.5\frac{nV_O}{V_{\text{in, max}}}$

$$B'_{max} = \frac{B_{max}}{2 - \frac{0.5nV_O}{V_{\text{in, max}}}} \tag{4.4}$$

Case 3: $D > 0.5$

$$B'_{max} = \frac{B_{max}}{1 + \frac{D-0.5}{D}} \tag{4.5}$$

In these cases are calculated in terms of the peak of the input voltage. One can

choose a higher input voltage for safety, and if the input voltage is less than the maximum, then one can have more leeway for the duty cycle to be used to operate in a chosen pattern. One can also set a maximum duty cycle $D_{\max}$ to be used in the circuit operation to limit the operation pattern. For the design method to be presented later, the D was set to be $D_{\max}$ to allow for leeway and flexibility in circuit operation.

4.3 Magnetizing Inductance

The magnetizing inductance $L_m = L_{m_1} = L_{m_2}$ is decided using Faraday's law. The Magnetizing Inductance is an important parameter related to the amount of energy that is stored in the primary side of the transformer before being released to the secondary side.

In this study, the magnetizing inductance is set to be the same for the two sets of transformers. The magnetizing inductance is then obtained from Faraday's Law using the peak primary side current. Assuming that the output power is limited by the efficiency of the converter, then $P_O = \eta P_{in}$, with η being the efficiency of the converter, and P_{in} being the input power to the converter. The peak primary side current $I_{\text{pri, pk}}$ is then as shown in Equation 4.6, using the output power P_O, efficiency η, and the amplitude of the input voltage $V_{\text{in, max}}$.

$$I_{\text{pri, pk}} = \frac{0.5 P_O}{\eta V_{\text{in, max}} D} \tag{4.6}$$

To assume and make sure that the converter operates in DCM, in the Faraday's Law calculation shown below, the current is set to go from zero to the peak primary side current in one cycle. The required magnetizing inductance is then shown in Equation 4.9. It is important to note that current rises and falls linearly in an inductor. Also, here, T_s represents the switching period of the converter while f_s represents the switching frequency.

$$V_{\text{in, min}} = L_m \frac{\Delta i_{L_m}}{\Delta t} \tag{4.7}$$

$$= L_m \frac{I_{\text{pri, pk}}}{D T_s} \tag{4.8}$$

$$L_m = \frac{V_{\text{in, max}} D}{I_{\text{pri, pk}} f_s} \tag{4.9}$$

4.4 Number of Turns for the Transformers

After deciding the magnetizing inductance, the number of turns for each transformer can be decided using the relation between flux and self inductance as shown in Equations 4.10 to 4.15. The number of turns is calculated using the saturation magnetic flux density for each transformer B'_{max}, since they operate separately, and the peak primary side current $I_{\text{pri, pk}}$. A geometric reference for the transformer is shown also in Figure 4.1. The cross-sectional areas A_{mo} for the outer legs and A_{mc} for the center leg can be obtained using the dimensions of the core chosen.

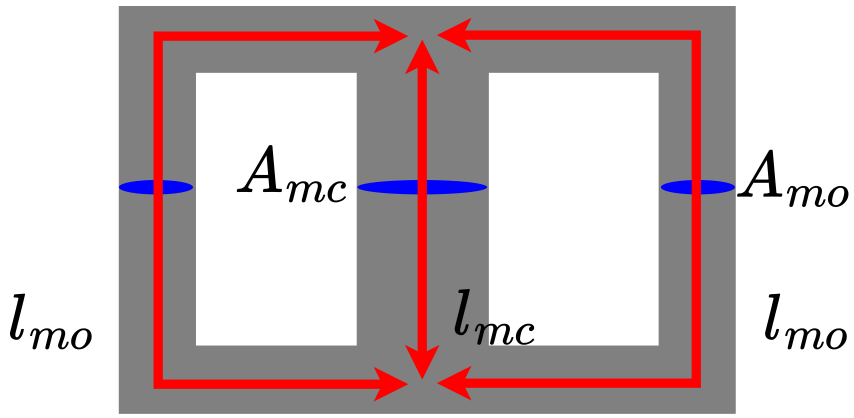


Figure 4.1: Geometric representation of the transformer. Here, the effective cross-sectional area is given as A_{mo} for the outer legs and A_{mc} for the center leg. The effective magnetic path length is given as l_{mo} for the outer legs and l_{mc} for the center leg.

For transformer $N_{12}N_{34}$ (outer leg transformer):

$$nN_{34}\Phi_{34} \geq L_m I_{\text{p, pk}} \tag{4.10}$$

$$N_{34} \geq \frac{L_m I_{\text{p, pk}}}{A_{mo}B'_{max}} \tag{4.11}$$

$$N_{12} = nN_{34} \tag{4.12}$$

For transformer $N_{56}N_{78}$ (center leg transformer):

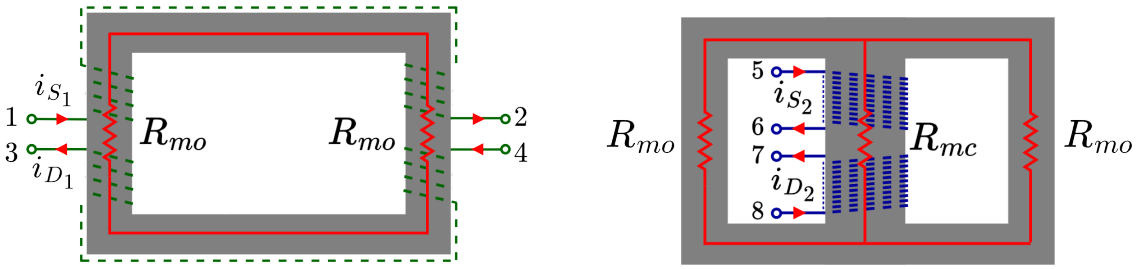
$$nN_{78}\Phi_{78} \geq L_m I_{\text{p, pk}} \tag{4.13}$$

$$N_{78} \geq \frac{L_m I_{\text{p, pk}}}{A_{mo}B'_{max}} \tag{4.14}$$

$$N_{56} = nN_{78} \tag{4.15}$$

4.5 Air Gap Design

An air gap is recommended to be inserted to avoid saturation of the core at any time. The air gap is going to be calculated based on the magnetic resistances. A representation of the magnetic resistances, R_{mo} for the outer legs and R_{mc} for the center leg, during the separate operation of the transformers is shown in Figure 4.2.



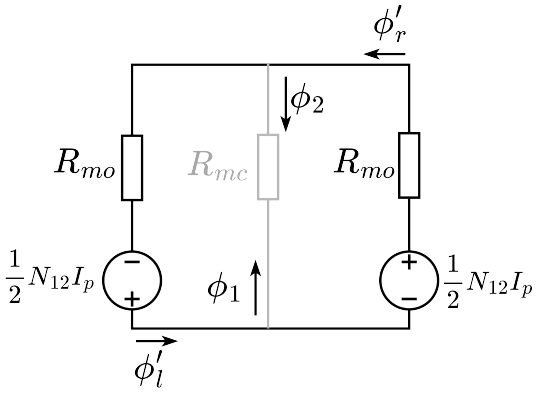
(a) Outer transformer. **(b)** Center transformer.

Figure 4.2: Magnetic resistances for the proposed transformer. Here, the magnetic resistance is given as R_{mo} for the outer legs and R_{mc} for the center leg.

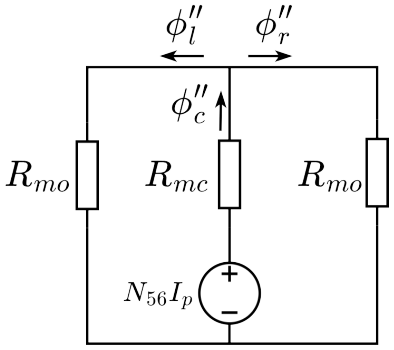
For the outer leg transformer, as have been discussed in Chapter 3, the flux in the middle leg is canceled, and thus the equivalent circuit is shown in Figure 4.3a, where it seems that there is no flux source in the middle leg. Moreover, since the magnetic resistances on the outer legs are equal, then the flux entering into the middle leg cancel each other out equally. Corresponding to the winding structure, the flux sources on the outer legs contain half the total number of turns wound on opposite polarity.

On the other hand, for the center transformer, as have been discussed, there is no net voltage produced on the outer legs, thus no flux sources are present there. Moreover, since the magnetic resistances on the outer legs are equal, then the flux entering the two outer legs should be equal to each other. The equivalent magnetic circuit is shown in Figure 4.3b

From the magnetic circuits in Figure 4.3, the expressions for the magnetic resistances can be derived.



(a) Outer transformer.



(b) Center transformer.

Figure 4.3: Equivalent magnetic circuits for the proposed transformer.

For the outer leg magnetic resistance, the following expression for magnetizing inductance can be derived from Figure 4.3a:

$$L_m = \frac{N_{12}^2}{2R_{mo}} \tag{4.16}$$

For the central leg magnetic resistance, the following expression for magnetizing inductance can be derived from Figure 4.3b:

$$L_m = \frac{N_{56}^2}{R_{mc} + 0.5R_{mo}} \tag{4.17}$$

The expressions for R_{mo} and R_{mc} can be derived using Equations 4.16 and 4.17:

$$R_{mo} = \frac{N_{12}^2}{2L_m} \tag{4.18}$$

$$R_{mc} = \frac{(2N_{56} - N_{12}) + (2N_{56} + N_{12})}{4L_m} \tag{4.19}$$

The air gap can then be calculated. An expression for the magnetic resistances can be calculated using the air gap length and magnetic path length. However, two cases can be considered for the air gaps in the outer and center legs, denoted as $l_{mo_{air}}$ and $l_{mc_{air}}$, respectively, and are described below. This is because of the

non-uniformity of the flux path in the core between the central and outer legs.

Firstly, it can happen that the air gap $l_{mo_{air}}$ in the outer legs is bigger than that of the central leg $l_{mc_{air}}$. The geometric representation of this case is shown in Figure 4.4. The resulting expressions for the outer and central leg resistance R_{mo} and R_{mc} in terms of the air gap length are shown in Equation 4.20. Using Equations 4.18 - 4.20 will result into the appropriate air gaps for this case shown in Equation 4.21. It is important that physical manipulation of the existing core by shaving the outer legs is needed to add the additional air gap for the outer legs, which can cause a challenge in manufacturability at the moment.

$$\begin{cases} R_{mo} = \dfrac{l_{mo} - l_{mo_{air}} + l_{mc_{air}}}{\mu_r \mu_0 A_{mo}} + \dfrac{l_{mo_{air}}}{\mu_0 A_{mo}} \\ R_{mc} = \dfrac{l_{mc}}{\mu_r \mu_0 A_{mc}} + \dfrac{l_{mc_{air}}}{\mu_0 A_{mc}} \end{cases} \tag{4.20}$$

$$\begin{cases} l_{mo_{air}} = \dfrac{\mu_0 \left(\mu_r A_{mo} R_{mo} - A_{mc} R_{mc}\right) + \frac{l_{mc}}{mu_r} - l_{mo}}{mu_r - 1} \\ l_{mc_{air}} = \dfrac{\mu_0 A_{mc} R_{mc}}{\mu_r - 1} - \dfrac{l_{mc}}{\mu_r} \end{cases} \tag{4.21}$$

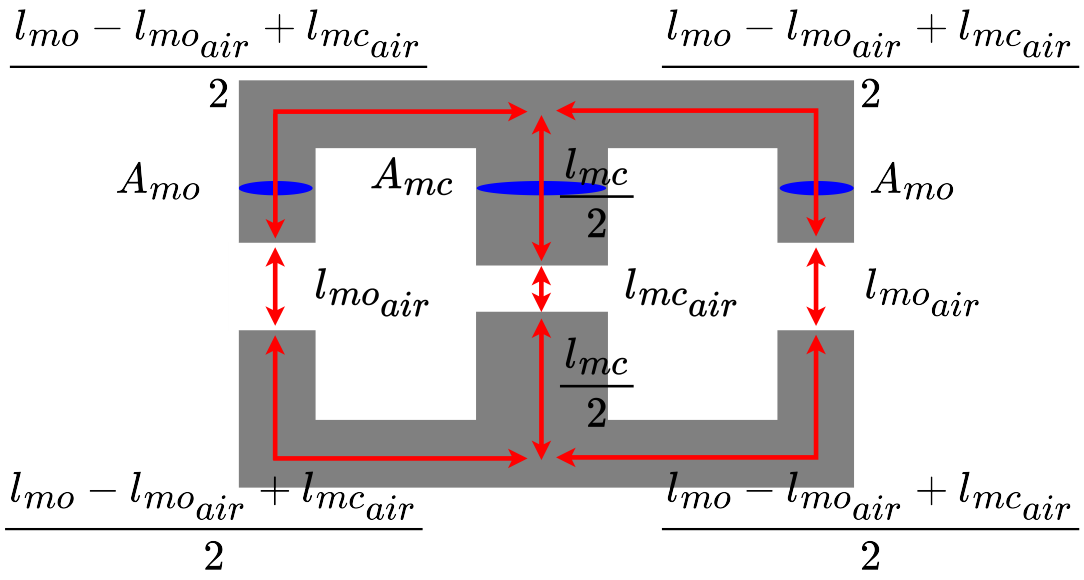


Figure 4.4: The air gaps of the proposed transformer, with $l_{mo_{air}} > l_{mc_{air}}$.

In the case that the air gap $l_{mc_{air}}$ in the central leg is bigger than that of the outer legs $l_{mo_{air}}$. The geometric representation of this case is shown in Figure 4.5. Similarly, the resulting expressions for the outer and central leg resistance R_{mo} and R_{mc} in terms of the air gap length are shown in Equation 4.22, and consequently, the appropriate air gaps for this case is shown in Equation 4.23. Shaving of the center leg is needed to add the additional air gap for the center legs.

$$\begin{cases} R_{mo} = \dfrac{l_{mo}}{\mu_r \mu_0 A_{mo}} + \dfrac{l_{mo_{air}}}{\mu_0 A_{mo}} \\ R_{mc} = \dfrac{l_{mc} - l_{mc_{air}} + l_{mo_{air}}}{\mu_r \mu_0 A_{mc}} + \dfrac{l_{mc_{air}}}{\mu_0 A_{mc}} \end{cases} \tag{4.22}$$

$$\begin{cases} l_{mo_{air}} = \dfrac{\mu_0 A_{mo} R_{mo}}{\mu_r - 1} - \dfrac{l_{mo}}{\mu_r} \\ l_{mc_{air}} = \dfrac{\mu_0 \left(\mu_r A_{mc} R_{mc} - A_{mo} R_{mo}\right) + \frac{l_{mo}}{mu_r} - l_{mc}}{mu_r - 1} \end{cases} \tag{4.23}$$

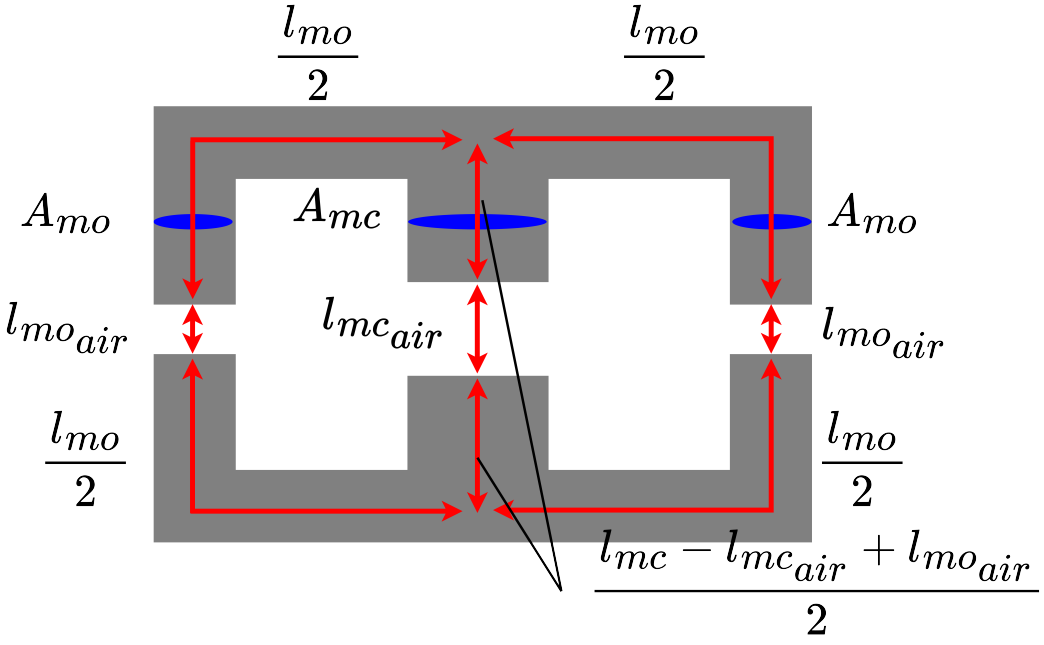


Figure 4.5: The air gaps of the proposed transformer, with $l_{mc_{air}} > l_{mo_{air}}$.

Chapter 5

Design Method Verification

This section describes experimental and simulation verification of the proposed transformer. Various waveforms and operational patterns will be shown in this section. For this study, the variation of the circuit operation Pattern 1 according to Chapter 3 was verified.

5.1 Prototype Circuit for Evaluation

A prototype two-phase interleaved flyback rectifier, shown in Figure **??**, was built to verify the integrated transformer.

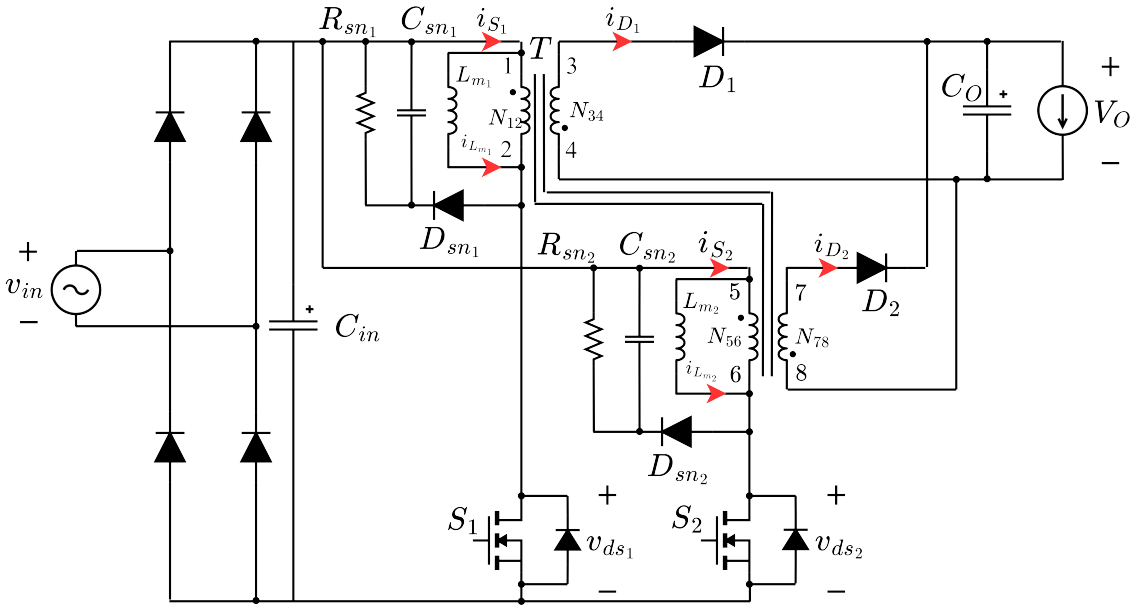


Figure 5.1: The prototype two-phase interleaved flyback PFC rectifier for the proposed non-coupled integrated transformer design evaluation.

The circuit was operated in discontinuous current mode, where the phases were driven with 180° difference. R_{sn_1} and R_{sn_2} are snubber resistors, C_{sn_1} and C_{sn_2} are snubber capacitors, and D_{sn_1} and D_{sn_2} are snubber diodes, which form a clamped RCD snubber circuit [25], [26]. This was added to protect the MOSFETs from surge voltages caused by transformer leakage inductance during turn-off of S_1 and S_2. Table **??** shows the circuit specifications and components used. The values for the snubber circuits are also fine-tuned using simulation. The prototype circuit has no control stage for the output voltage yet since this study only focuses on a proof of concept first.

Table 5.1: Operating Conditions for the two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer.

Parameter	Symbol	Value
AC input voltage	v_{in}	100 V V_{rms}
Maximum Rated Power	$P_{O,\text{ max}}$	240 W
Output voltage	V_O	48 V
Switching frequency	f_s	100 kHz
Line frequency	f_{line}	60 Hz
Voltage Ripple	ΔV	8 V
Target Efficiency	η	90%

Shown in Table **??** are the other circuit components used to accommodate the operating conditions of the two-phase interleaved flyback PFC rectifier.

Table 5.2: Circuit components used in the two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer.

Component	**Symbol**	**Value**
Bridge diode	BD	CD-DF408S
Input capacitor	C_{in}	1 μF
Switch	S_1, S_2	STF15N80k5
Secondary diode	D_1, D_2	STPSC30H12CWL
Output capacitor	C_o	2 x 1 mF (parallel)
Snubber capacitor	C_{sn_1}, C_{sn_2}	2.2 nF
Snubber resistor	R_{sn_1}, R_{sn_2}	30 kΩ
Snubber diode	D_{sn_1}, D_{sn_2}	VS-E7MH0112-M3/I

To design the output capacitor, the following formula was used and it depends on the output voltage ripple [27].

$$C_{\mathrm{O}} \geq \frac{P_O}{2\pi f_{\mathrm{line}} \Delta V V_O} \tag{5.1}$$

Finally, shown in Figure **??** is the prototype evaluation circuit with the proposed transformer used in the experiment.

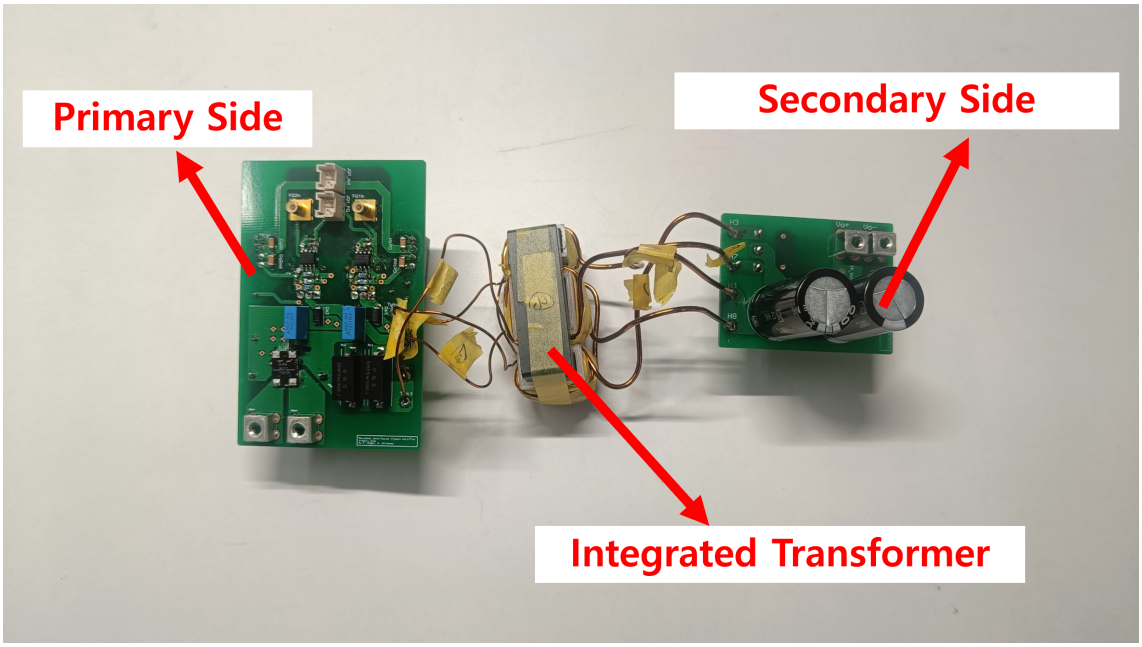


Figure 5.2: Actual prototype two-phase interleaved flyback PFC rectifier for the proposed non-coupled integrated transformer design evaluation.

5.2 Prototype Transformer Design and Size Reduction

A prototype for the transformer was built. The specifications are shown in Table ??.

The design process for this transformer is performed according to the method described in Chapter 4. The desired operating conditions for the converter are shown in Table **??**.

A switch with maximum V_{DS} at 800 V was selected, with a margin V_{margin} and spike voltage V_{spike} of 100 V each. After using Equation 4.3, a turns ratio n was selected to be 3. In the computation of Equation 4.3, a maximum input voltage of 240 V V_{rms} was assumed for safety. Afterwards, with the operation pattern of Pattern 1, the maximum duty was assumed to be 0.47, which will then provide the saturation magnetic flux density to be the same as that of the core specification. The magnetizing inductance required to support the maximum output power is then

82.8 μH, obtained using Equation 4.6 and 4.9, with the peak primary side current $I_{\text{pri, pk}}$ as 8.02 A.

A core bigger than the expected area product method was first chosen. Given what was available in the laboratory, the core OP47721EC by Magnetics, inc. was selected as the core for this study. The area product of the core according to geometrical specifications is 46349 mm^4, and thus a suitable core was selected. The expected area product according to circuit specifications is shown in Table **??**. Using the equations described in Chapter 3, a 19% volume reduction is expected.

Table 5.3: Volume analysis of the conventional and proposed transformer using the chosen core OP44721EC.

Item	**Symbol**	**Conventional**	**Integrated**
Magnetizing inductance	L_m	82.8 μH	
Turns ratio	n	3	
Primary side peak current	$I_{\text{p,pk}}$	8.02 A	
Primary side rms current	$I_{\text{p,rms}}$	2.25 A	
Secondary rms current	$I_{\text{s,rms}}$	6.88 A	
Maximum flux density	B_{max}	0.3 T	
Maximum current density	J_{max}	5 A/mm^2	
Occupancy ratio	K_u	0.3	
Area product (circuit spec.)	AP	6703.21 mm^4	13406.42 mm^4
Core volume	Vol.	10313.57 mm^3	8672.64 mm^3

Finally, a number of turns was selected. The number of turns were fine-tuned according to the inductance measured using the LCR Meter, until the magnetizing inductance of the transformers were reached. This then led to N_{12} to have 24 turns, N_{34} to have 8 turns, N_{56} to have 15 turns, and N_{78} to have 5 turns. The specifications of the transformer prototype are shown in Table **??**, and its diagram representation is shown in Figure **??**. The actual prototype manufactured is shown in Figure **??**.

Table 5.4: Specifications of the prototype non-coupled integrated transformer.

Item	**Symbol**	**Value, Component**
Magnetic core	T	OP47721EC
Relative Permeability	μ_r	2500
Mag. path length (outer leg)	l_{mo}	71.4 mm
Mag. path length (center leg)	l_{mc}	31.7 mm
Cross-sectional Area (outer leg)	A_{mo}	117.6 mm^2
Cross-sectional Area (center leg)	A_{mc}	243.4 mm^2
Mag. inductance (Target)	L_m	82.8 μH
Mag. inductance of N_{12}, N_{34}	L_{m_1}	86.1086 μH
Mag. inductance of N_{56}, N_{78}	L_{m_2}	75.0598 μH
Turns ratio	n	3
Magnetic core	T	OP44721EC
Number of turns (outer primary)	N_{12}	24
Number of turns (outer secondary)	N_{34}	8
Number of turns (center primary)	N_{56}	15
Number of turns (center secondary)	N_{78}	5
Air gap (outer leg)	l_{go}	0.50 mm
Air gap (center leg)	l_{gc}	0.65 mm
Coupling coeff. of transformer N_{12}, N_{34}	$k_{N_{12}}, k_{N_{34}}$	0.9701
Coupling coeff. of transformer N_{56}, N_{78}	$k_{N_{56}}, k_{N_{78}}$	0.9787
Coupling coeff. between N_{12}, N_{34} and N_{56}, N_{78}	$k_{N_{12}}, k_{N_{56}}$	0.00053

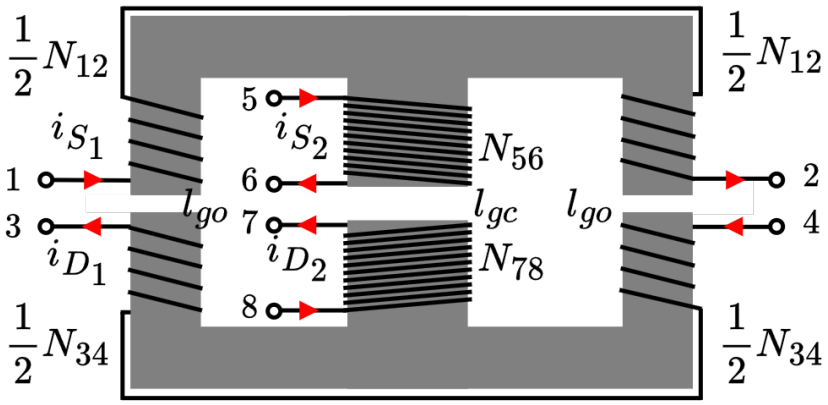


Figure 5.3: Diagram representation of the proposed non-coupled integrated transformer.

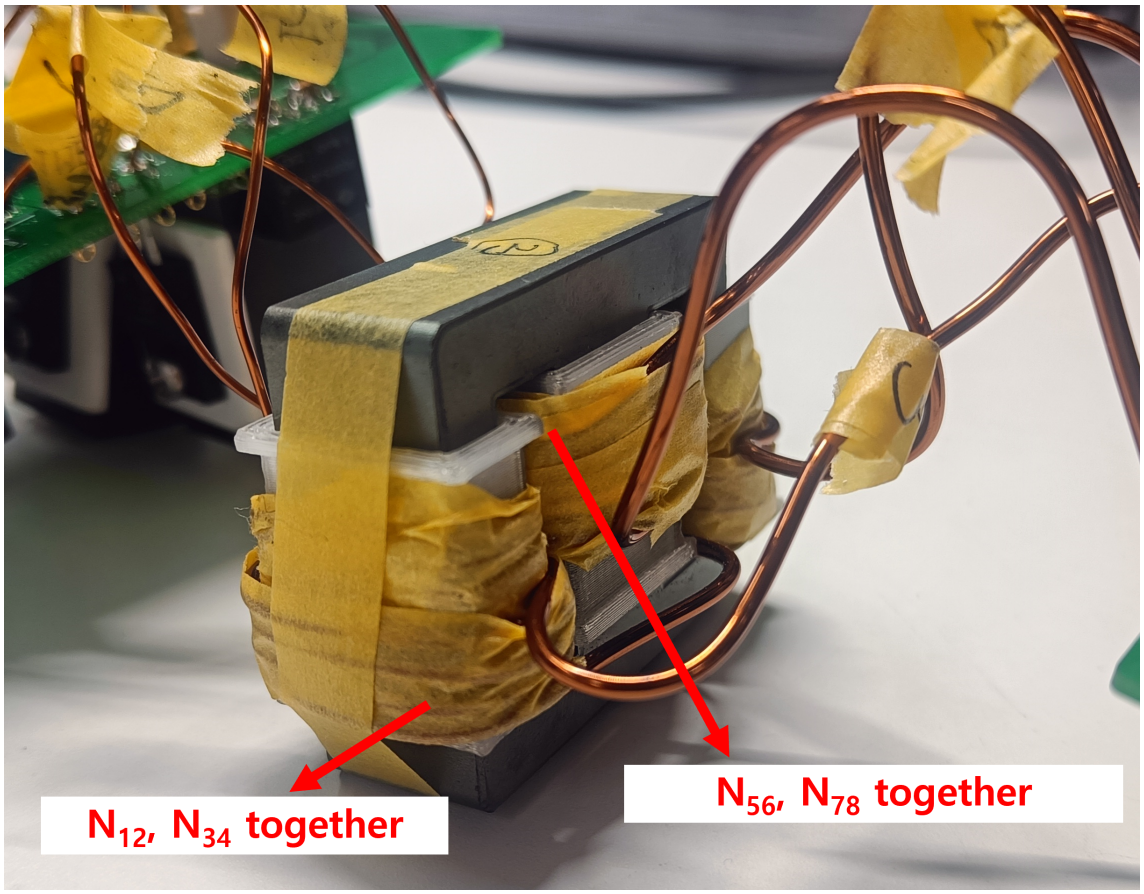


Figure 5.4: Actual prototype non-coupled integrated transformer used for design method verification.

The coupling coefficient measurement method is described as below. To take into account as much as possible the other parasitic inductances along the path of the inductors of the transformer, the following method is performed.

The involved inductors are first measured. Afterwards, they are connected in either series-aiding and series-opposing connection. This method aims to obtain the mutual inductance between the inductors.

For the series-aiding configuration measurement shown in Figure **??**, the total inductance $L_{meas\text{: series, aid}}$ measured is:

$$L_{meas\text{: series, aid}} = L_1 + L_2 + 2M \tag{5.2}$$

For the series-opposing configuration measurement shown in Figure **??**, the total inductance $L_{meas\text{: series, oppose}}$ is measured as:

$$L_{meas\text{: series, oppose}} = L_1 + L_2 - 2M \tag{5.3}$$

From equation **??** and **??**, we can obtain the mutual inductance M as:

$$M = \frac{L_{meas\text{: series, aid}} + L_{meas\text{: series, oppose}}}{4} \tag{5.4}$$

And from here, one can obtain the coupling coefficient k between the two involved inductors L_1, L_2 as:

$$k = \frac{M}{\sqrt{L_1 L_2}} \tag{5.5}$$

And lastly, the magnetizing inductance L_m of the transformer being examined is obtained as:

$$L_m = k^2 L \tag{5.6}$$

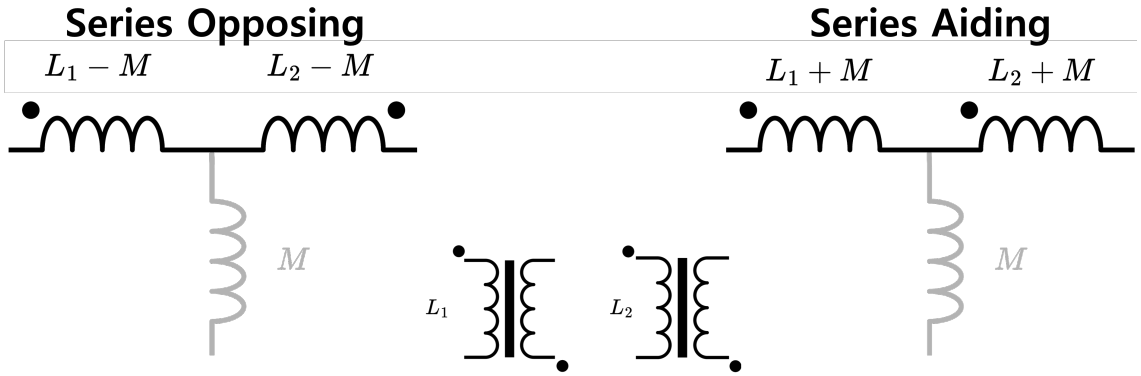


Figure 5.5: Transformer coupling coefficient measurement method.

The equipment used all throughout the experimentation phase are listed in Table ??.

Table 5.5: Various measuring equipment used in the experimental verification of the two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer.

Equipment	**Model**
LCR Meter	Hioki IM3536
Electronic Load	Kikusui PLZ1004WH
Power Analyzer	Hioki PW8001
AC Power Supply	Kikusui CR2000M
Function Generator	Tektronix AFG31000 Series
8-ch. Oscilloscope	Tektronix MSO58
3.3 V Gate Driver Supply	Kikusui PWR401L
5V Converter Power Supply	Matsusada TB160V15A720W
Power Analyzer Probes (Current)	Hioki CT6843A AC/DC Current Probe
Voltage Probes	THDP0200
Current Probes	TCP0030A

5.3 Simulation Verification

Before creating the actual prototype, a verification through simulation was first performed. The prototype circuit shown first in Figure **??** was made in PLECS. The resulting circuit is shown in Figure **??**.

The simulation parameters are listed in Table **??** for the circuit parameters and Table **??** for the transformer parameters. During the calculation stage, the number of turns of N_{56} is 18 and of N_{78} is 6.

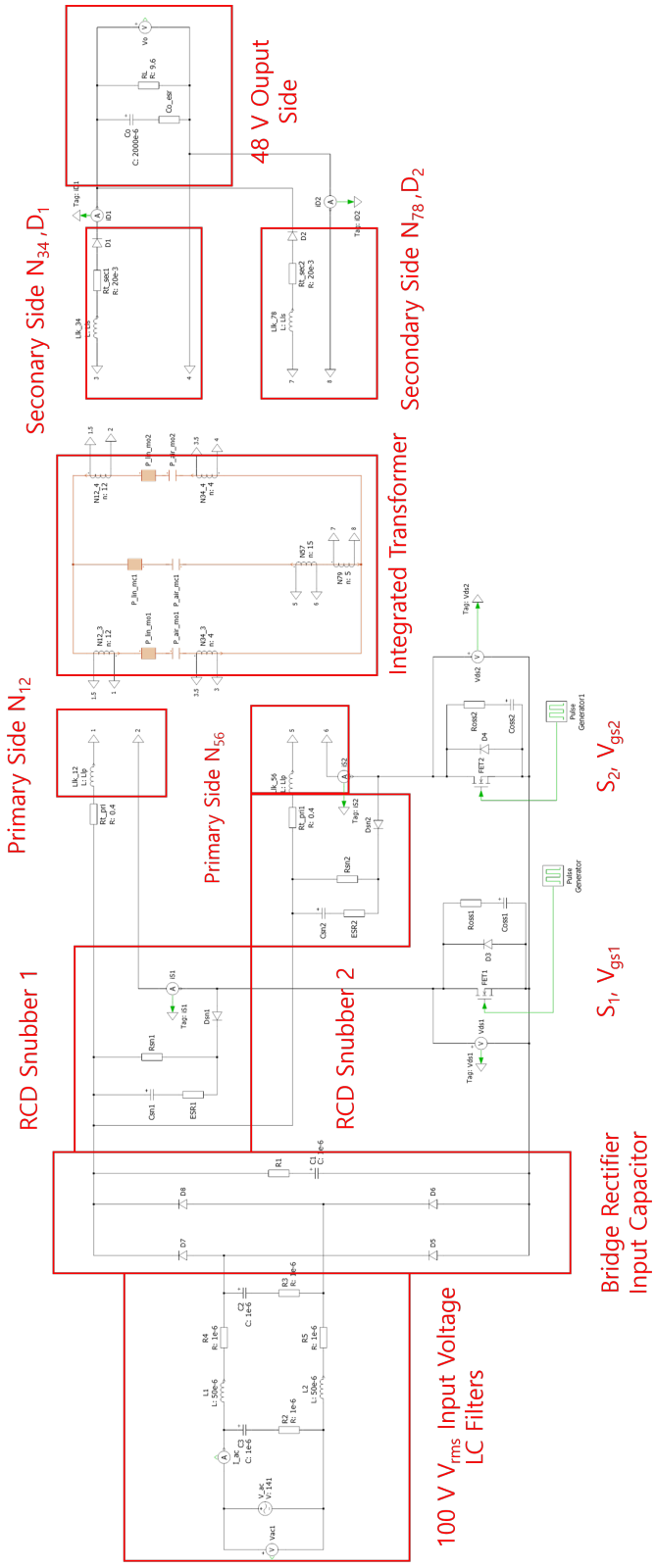


Figure 5.6: Simulation circuit for the two-phase interleaved flyback recitfier with the proposed non-coupled integrated transformer in PLECS.

Table 5.6: Simulation parameters for the two-phase interleaved flyback PFC rectifier.

Item	**Symbol**	**Value, Component**
AC input voltage	v_{in}	100 V V_{rms}
Max. Rated Power	P_o	240 W
Output voltage	V_o	48 V
Switching frequency	f_s	100 kHz
Line frequency	f_{line}	60 Hz
Input capacitor	C_{in}	1 μF
Output capacitor	C_o	2 x 1 mF (parallel)
Snubber capacitor	C_{sn1}, C_{sn2}	2.2 nF
Snubber resistor	R_{sn1}, R_{sn2}	30 kΩ

Table 5.7: Simulation parameters for the proposed non-coupled integrated transfomer.

Item	**Symbol**	**Value, Component**
Magnetic core	T	OP47721EC
Relative Permeability	μ_r	2500
Mag. path length (outer leg)	l_{mo}	71.4 mm
Mag. path length (center leg)	l_{mc}	31.7 mm
Cross-sectional Area (outer leg)	A_{mo}	117.6 mm^2
Cross-sectional Area (center leg)	A_{mc}	243.4 mm^2
Mag. inductance	L_m	82.8 μH
Mag. inductance of N_{56}, N_{78}	L_{m_2}	75.0598 μH
Turns ratio	n	3
Magnetic core	T	OP44721EC
Number of turns (outer primary)	N_{12}	24
Number of turns (outer secondary)	N_{34}	8
Number of turns (center primary)	N_{56}	18
Number of turns (center secondary)	N_{78}	6
Air gap (outer leg)	l_{go}	0.50 mm
Air gap (center leg)	l_{gc}	0.65 mm
Coupling coeff. of transformer N_{12}, N_{34}	$k_{N_{12}}, k_{N_{34}}$	0.9990
Coupling coeff. of transformer N_{56}, N_{78}	$k_{N_{56}}, k_{N_{78}}$	0.9990
Coupling coeff. between N_{12}, N_{34} and N_{56}, N_{78}	$k_{N_{12}}, k_{N_{56}}$	0.0000

5.3.1 Flux Operation

The magnetic component designed in PLECS is shown in Figure **??**.

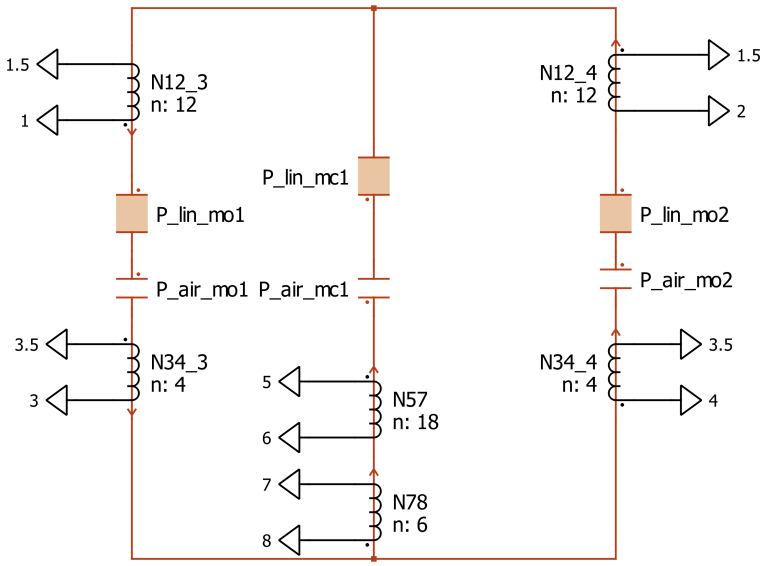
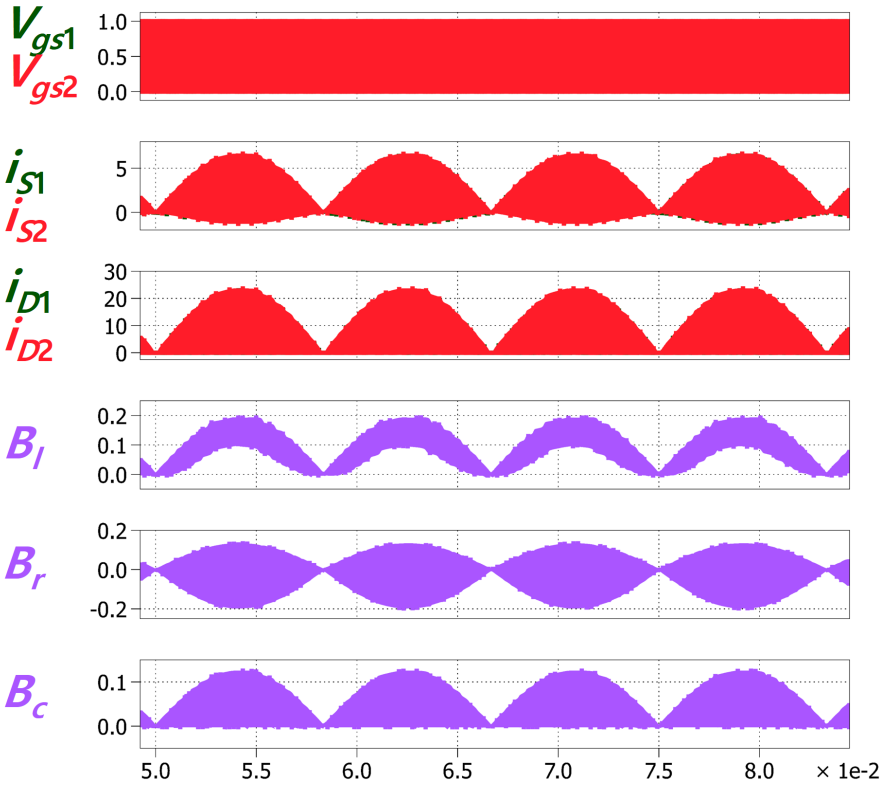
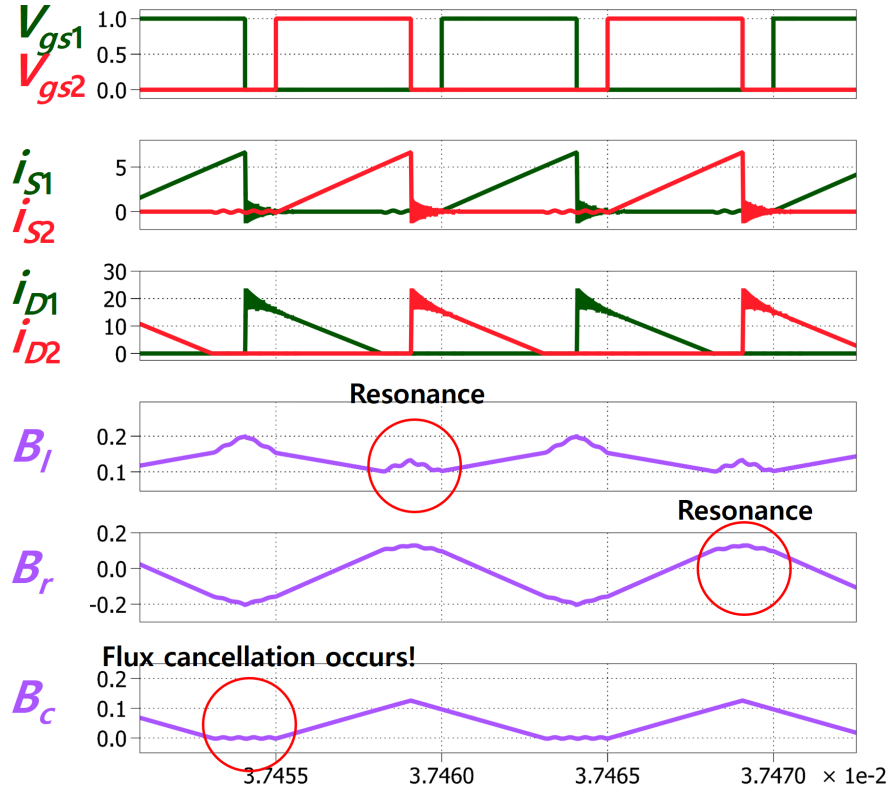


Figure 5.7: The proposed non-coupled integrated transformer in PLECS.

The waveforms obtained from the magnetic component designed is shown in Figure **??**, with a zoomed in version in Figure **??**. For Pattern 1, a somewhat similar shape can be observed to that of the theoretical analysis in Chapter 3 (see Figure 3.13b). Flux cancellation does not occur in any of the left and right outer legs, but flux cancellation is seen in the center leg. However, contrary to the analysis in Chapter 3, the peak flux is instead seen in the left leg instead. The differences and irregularities in the shape arise due to the unequal magnetic inductances present in the transformer. In the particular case shown in the figure, $L_{m_1} < L_{m_2}$ because L_{m_1} reaches a higher peak i_{S_1}. This causes the flux during the operation of N_{12} N_{34} to rise and fall from a higher peak, which then causes the differences in the expected result theoretically. Although it was designed by simulation, the method used to create the magnetic element makes it hard to make the inductances equal.



(a) Zoomed-out flux density waveforms.



(b) Zoomed-in flux density waveforms.

Figure 5.8: Simulated flux density waveforms for the proposed transformer.

Nevertheless, flux cancellation occurs in the center leg, and also, the changes in magnetic flux are linear, except at the peaks, which are due to the resonance present between the leakage inductance of the transformers and the switch capacitances, which then implies that no coupling occurs between the two transformers and therefore operate separately from each other during each phase.

To account for this non-ideal operation, the magnetic component has been changed to the component shown in Figure **??**, where the transformer is assumed to be one big coupled inductor composed of four different magnetic components. This time, it is ensured that the magnetizing inductances of the transformers are equal and will be used to simulate the remaining cases as will be described in the next subsections.

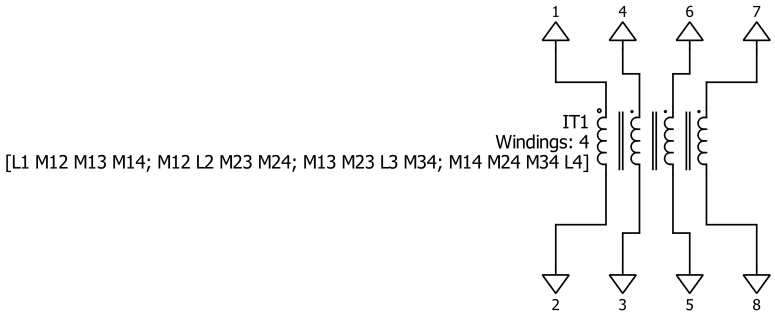
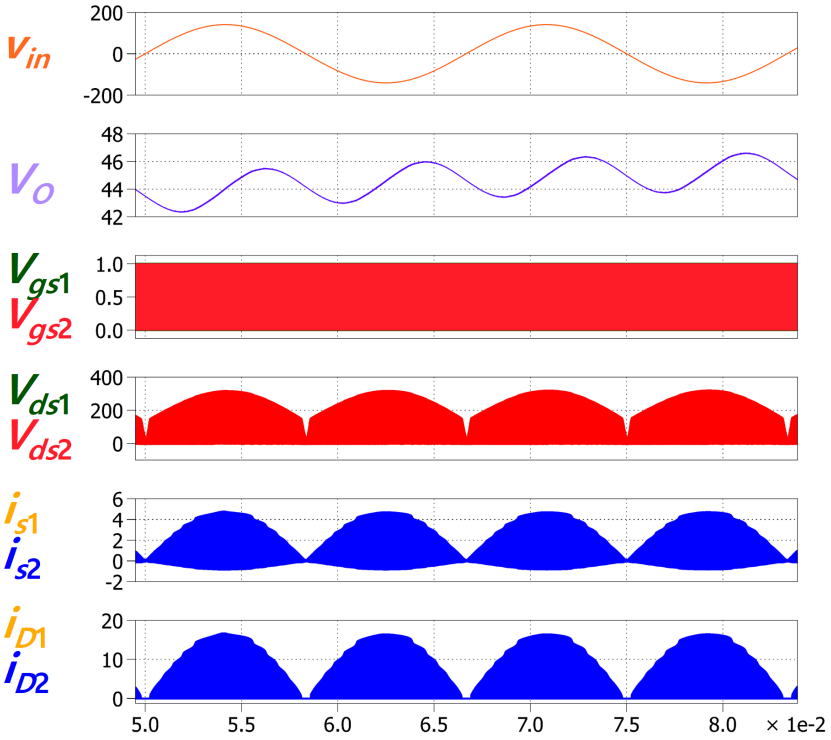


Figure 5.9: New magnetic model for the proposed non-coupled integrated transformer used in simulation.

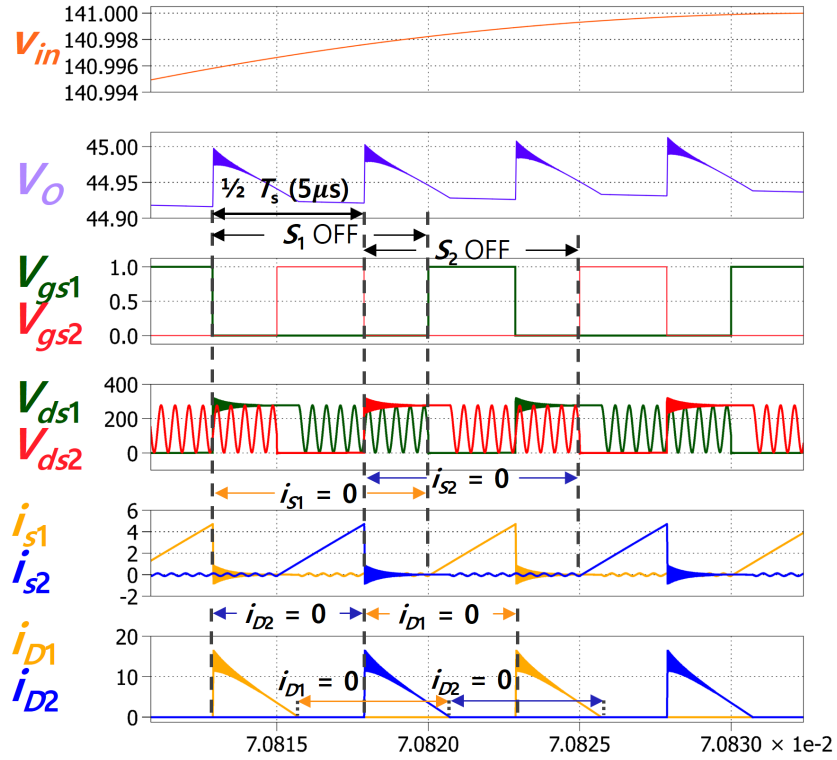
5.3.2 Two-phase Operation

Figure **??** shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 100 W. Proper PFC Operation can be observed from Figure **??**, especially that the current through the switches follow the same phase as of the rectified voltage. On the other hand, from the zoomed-in wave forms in Figure **??**, wherein during off-time of S_1 and S_2, the current i_{s_1} is zero during the off-time of S_1 and i_{s_2} is zero during the off-time of S_2. No current is induced in either switch except during the resonance between the leakage inductance and the parasitic capacitance of the switch. During the inactive period of the transformer, for example, between the start of the off-time of S_1 and the start of the off-time of S_2, the secondary diode side current is zero. Moreover, after i_{D1} and i_{D2} reached zero when S_1 and S_2 switched off, the currents remained at zero. Lastly, a linear increase and decrease of i_{S_1}, i_{S_2}, i_{D_1}, and i_{D2} was observed throughout the circuit operation. Therefore, from simulation, the two flyback converters N_{12}, N_{34}, S_1, D_1, C_O, V_O and N_{56}, N_{78}, S_2, D_2, C_O, V_O can operate independently without magnetic coupling interference, as coupling was eliminated between the flyback transformers.

The same phenomenon was observed at low (10 W) and high loads (240 W), which therefore implies that the transformer can operate separately in two phases at any load with out any magnetic interference. The results are shown in Figures **??** and **??** respectively.

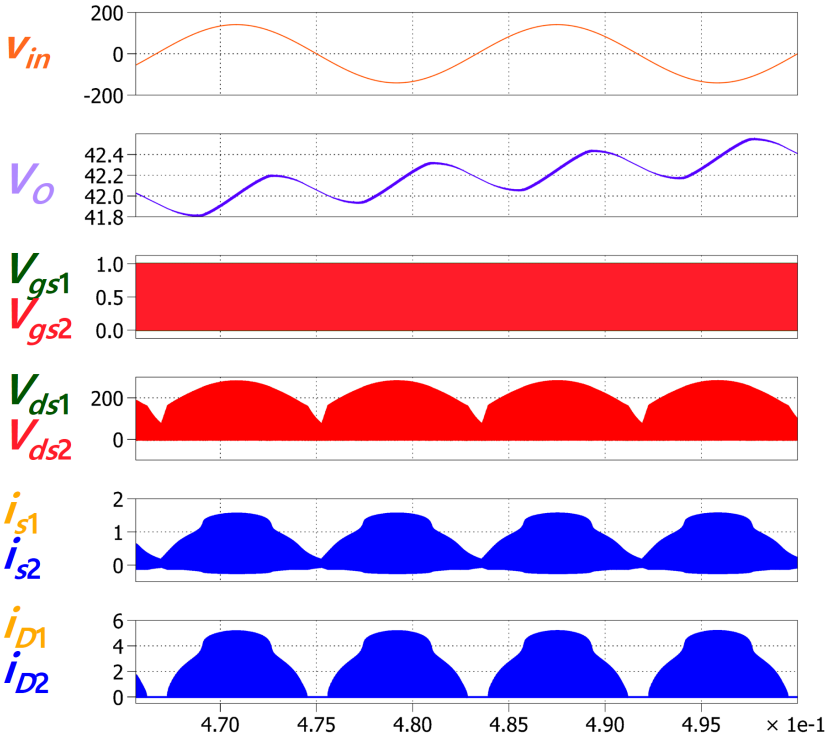


(a) Zoomed-out simulated waveforms.

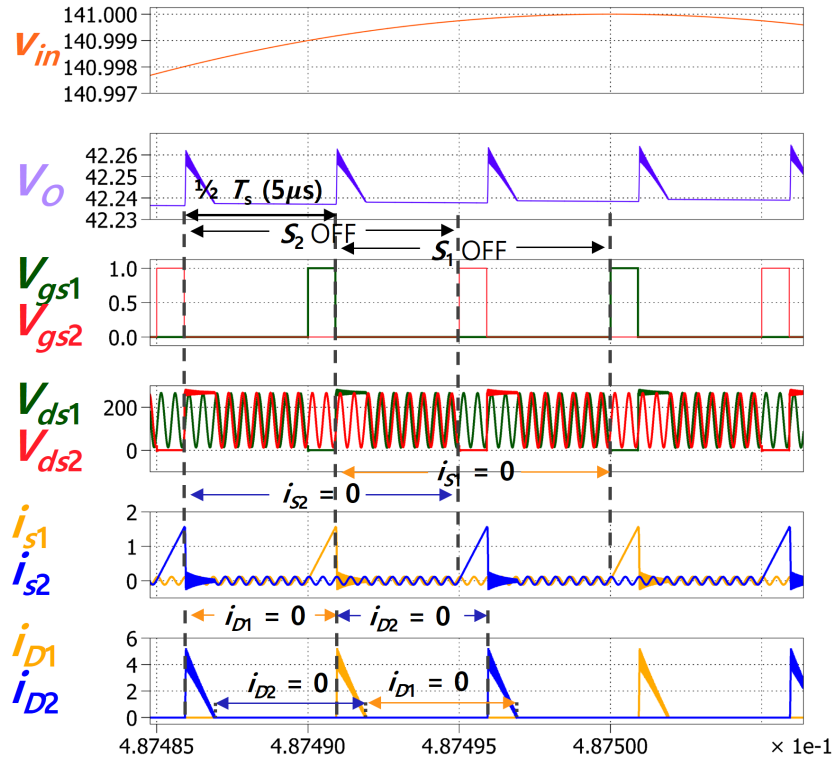


(b) Zoomed-out simulated waveforms.

Figure 5.10: Simulated waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 100 W, two-phase AC drive.

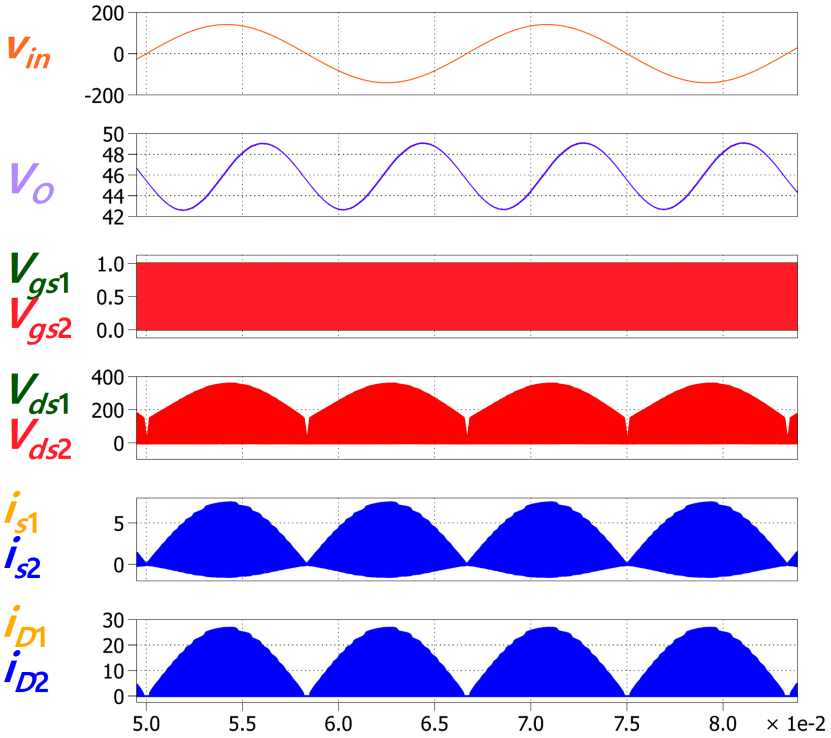


(a) Zoomed-out simulated waveforms.

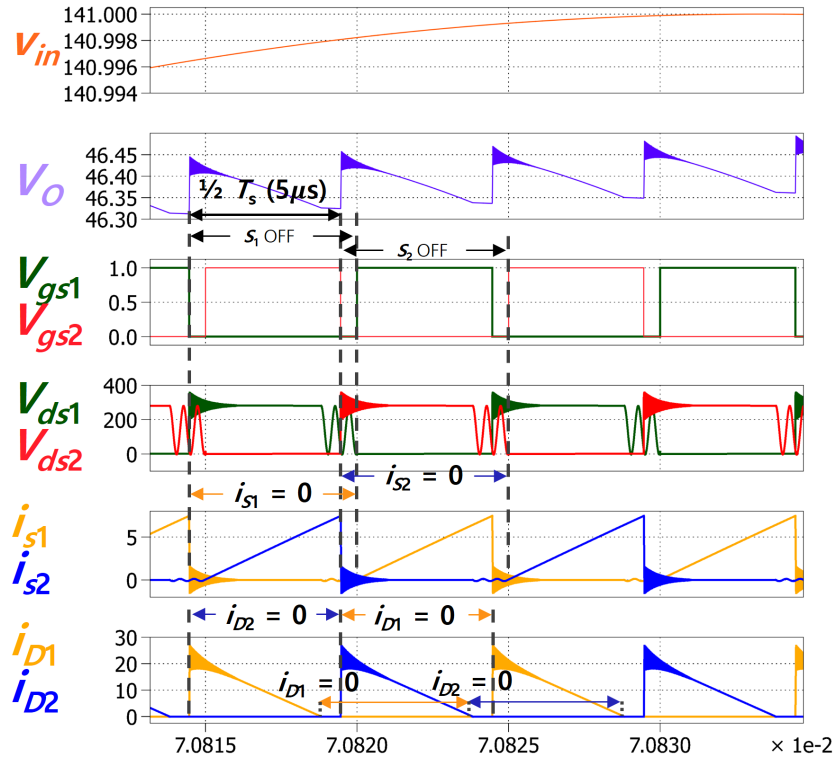


(b) Zoomed-in Simulated Waveforms.

Figure 5.11: Simulated waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 10 W, two-phase AC drive.



(a) Zoomed-out simulated waveforms.

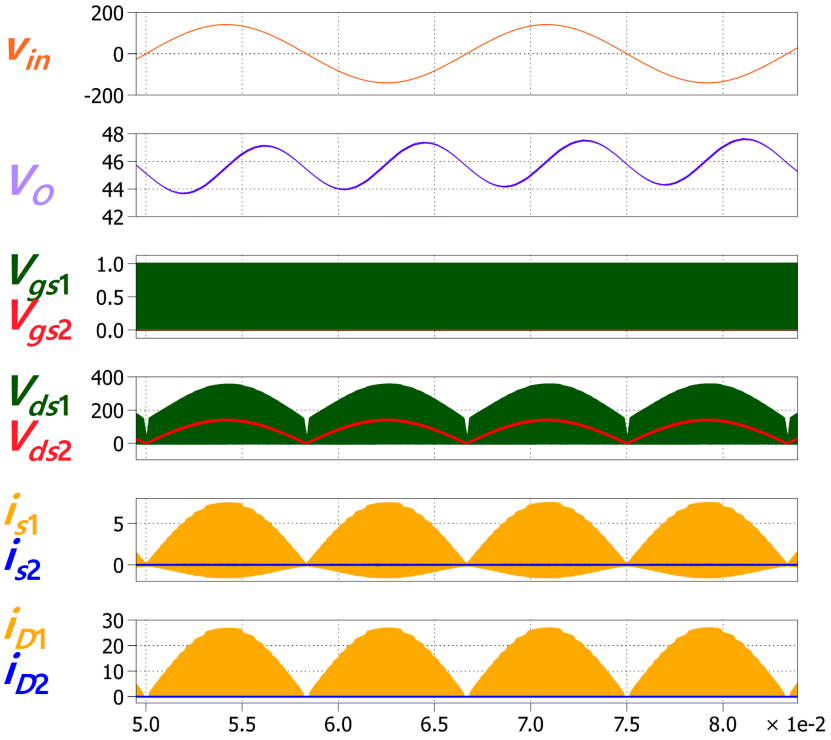


(b) Zoomed-in simulated waveforms.

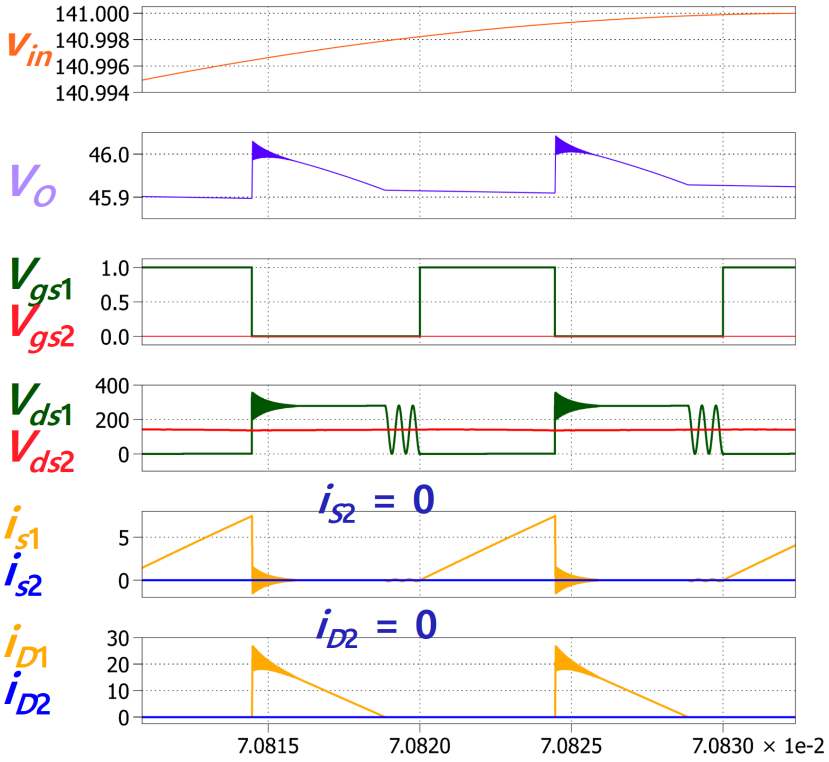
Figure 5.12: Simulated waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 240 W, two-phase AC drive.

5.3.3 Single-phase Operation

Single-phase drive at lower power is also feasible with this transformer. Figure **??** shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 120 W AC, with only single-phase drive using S_1. PFC operation can still be observed from the zoomed-out view of the waveforms from Figure **??**. Moreover, from the zoomed-in waveforms shown in Figure **??**, i_{S_2} and i_{D_2} are zero all throughout the circuit operation. This then confirms that only the flyback converter N_{12}, N_{34}, S_1, D_1, C_O, and V_O is operating. There is a voltage seen at V_{DS_2} since the two flyback converters are in parallel, although the switch is kept off. Therefore, from simulation, non-influence of the inactive transformer to the active one is confirmed, due to the elimination of coupling between the two transformers.



(a) Zoomed-out simulated waveforms.

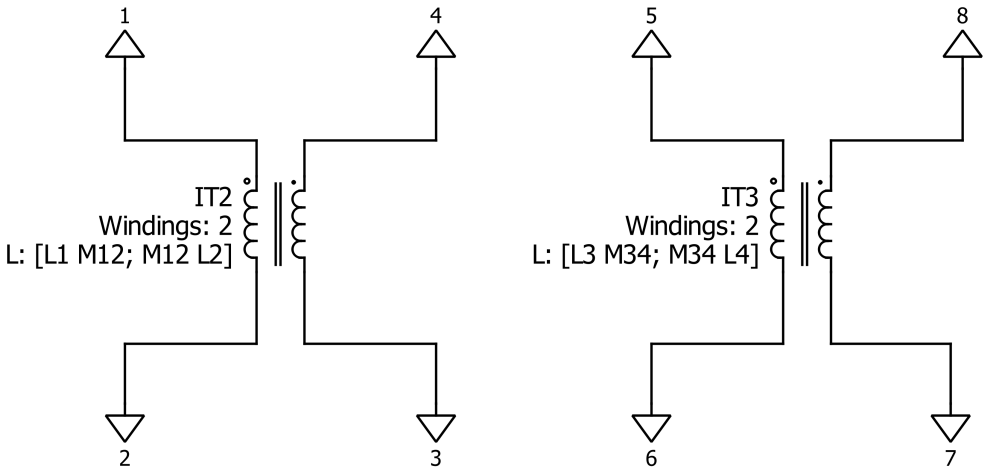


(b) Zoomed-in simulated waveforms.

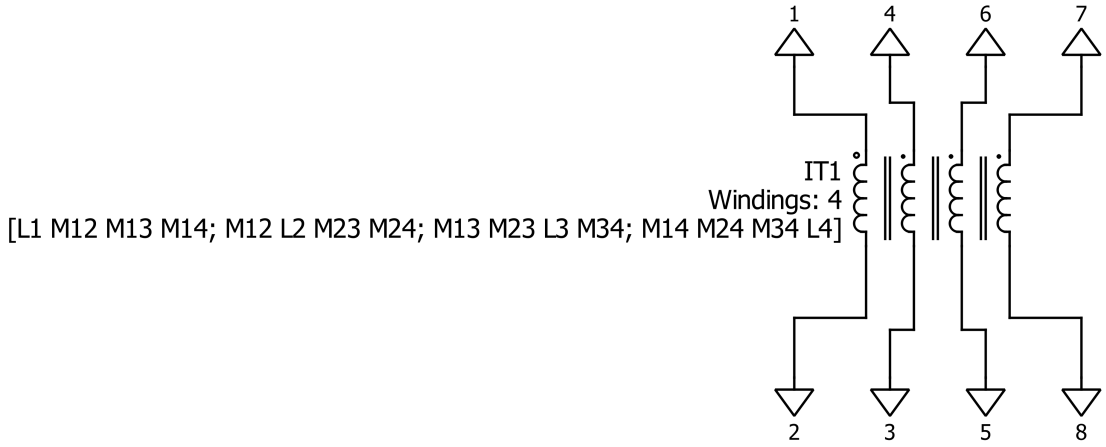
Figure 5.13: Simulated waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 120 W, one-phase AC drive.

5.3.4 Comparison of Efficiency between the Circuit with the Conventional and Proposed Transformer

To see if the integration of the transformer into one core has an effect on conversion efficiency, an efficiency analysis has been performed using PLECS across the power range 10 W - 240 W, at 100 V V_{rms}, 100 kHz switching frequency. Both the conventional and proposed simulated transformer are put into the circuit and then once at the steady state, the input and output power have been calculated, from where the efficiency $\left(\eta = \frac{P_O}{P_{in}}\right)$ can be obtained. The magnetic models used for the simulation are shown in Figure **??**.



(a) Conventional transformer in PLECS.



(b) Proposed transformer in PLECS.

Figure 5.14: Magnetic elements used in the efficiency comparison.

The results of the analysis are shown in Figure **??**. As seen, there is almost no difference between the efficiency of the converter when using the conventional and proposed transformer. Therefore, the integration into one core will not affect the efficiency too much, and it is better to use the integrated magnetic component instead, to allow for circuit miniaturization and reduction of construction cost.

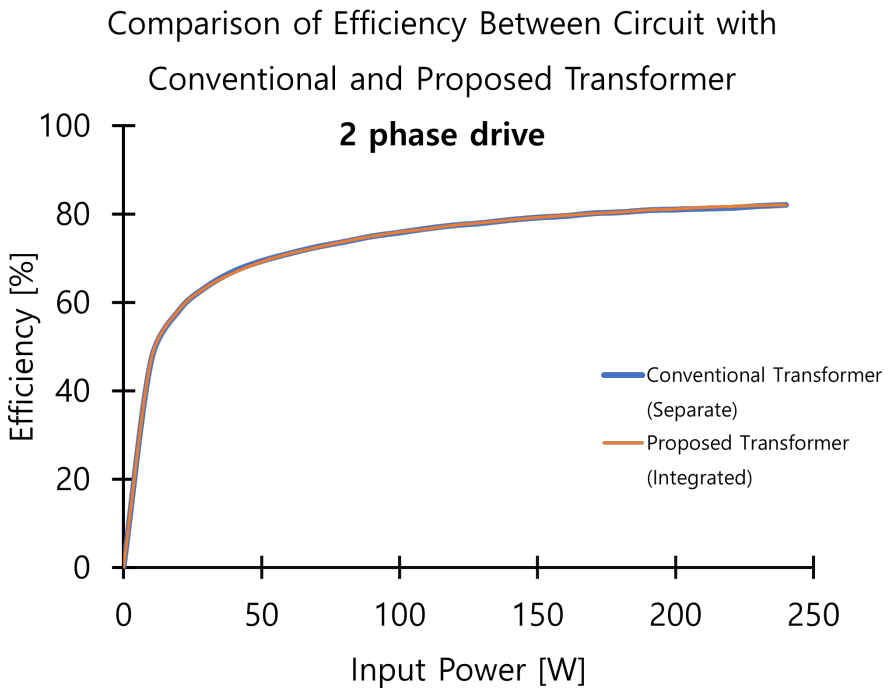
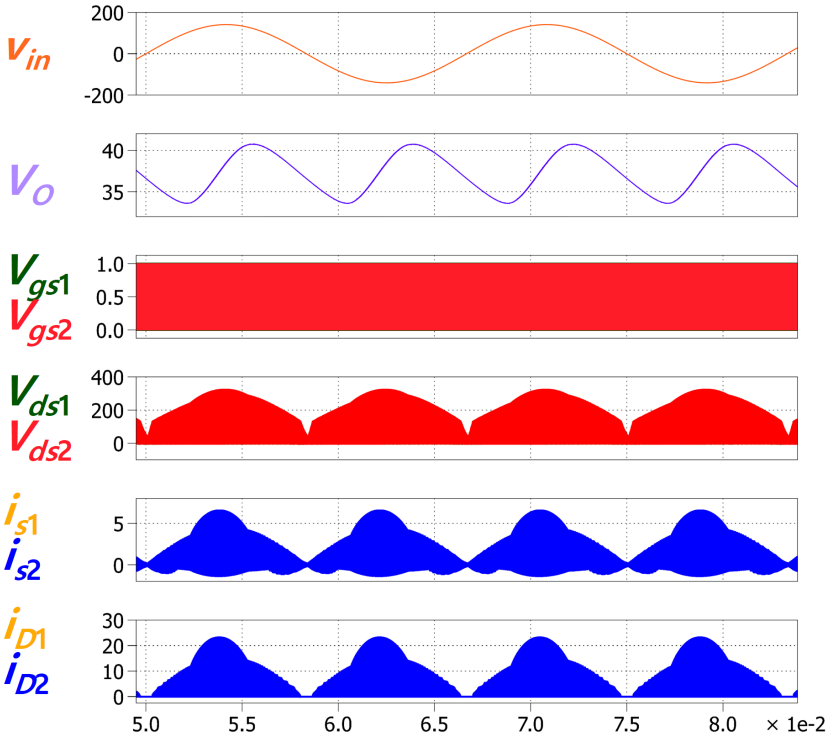


Figure 5.15: Comparison of converter efficiency between circuit with conventional and proposed transformer (2-phase drive).

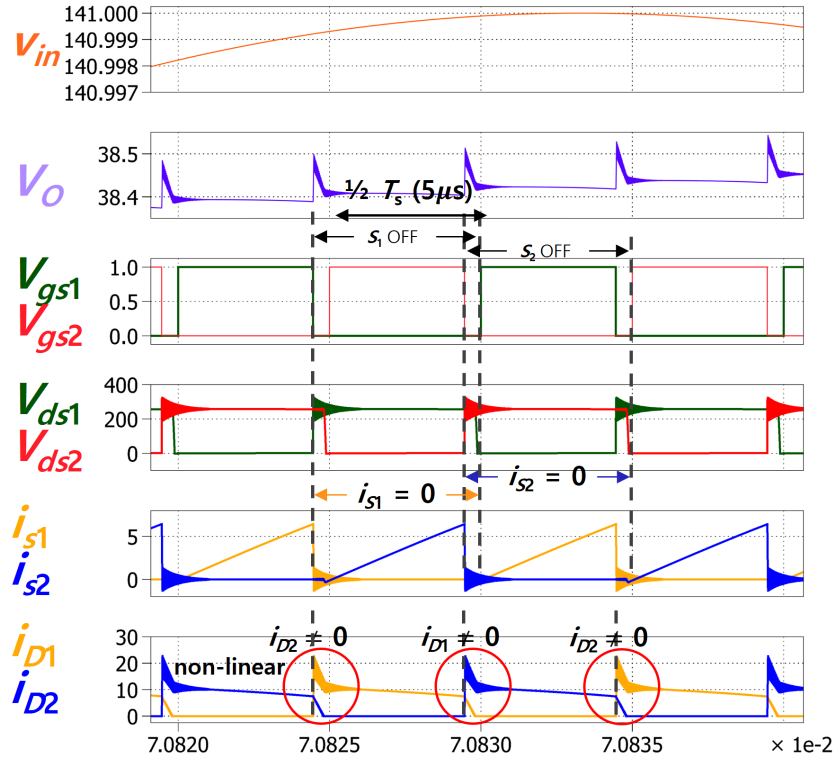
However though, a detailed loss analysis of the core model has not been performed yet using Finite Element Analysis (FEM) methods and calculation using excitation. Especially that the operation of this transformer is special, then the losses that can be obtained from here may not be straightforward compared to that of traditional analysis methods. Therefore, this part is going to be investigated further and results will be used to optimize the design of the proposed transformer in this study.

5.3.5 Investigation of the Effect of Coupling to Circuit Operation

The effect of coupling to the operation of the circuit has been observed through simulation. It is still difficult as of now to obtain the operational principle theoretically. Figure **??** shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 240 W, with coupling coefficient between the two transformers, also known as $k_{N_{12}N_{34},N_{56}N_{78}}$ set to 0.8. Proper PFC Operation can be observed from Figure **??**, however, there are strange peaks seen throughout the waveform, especially at peak input voltage and regions near the lowest input voltage, implying a possible interaction between the operation in two phases. From the zoomed-in wave forms in Figure **??**, it is clearly observed that the transformers do not operate independently of each other in each phase. The decrease of i_D is not linear to zero, and overlaps a bit with the discharge of the other transformer, as labeled in the figure. Therefore, the two flyback converters N_{12}, N_{34}, S_1, D_1, C_O, V_O and N_{56}, N_{78}, S_2, D_2, C_O, V_O cannot operate independently due to magnetic coupling interference provided by coupling between the flyback transformers.



(a) Zoomed-out simulated waveforms.



(b) Zoomed-in simulated waveforms.

Figure 5.16: Simulated waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 120 W, one-phase AC drive, with $k_{N_{12}N_{34},N_{56}N_{78}} = 0.8$.

An analysis of the efficiency at 100 W with the presence of coupling was performed. The coupling coefficient between the two transformers $k_{N_{12}N_{34},N_{56}N_{78}}$ was changed from $0.0 - 1.0$. From Figure **??**, the presence of coupling has a negative effect on the efficiency of the transformer, therefore necessitating the need to have the transformers operate completely independent of each other by elimination of coupling, to make power conversion as efficient as possible.

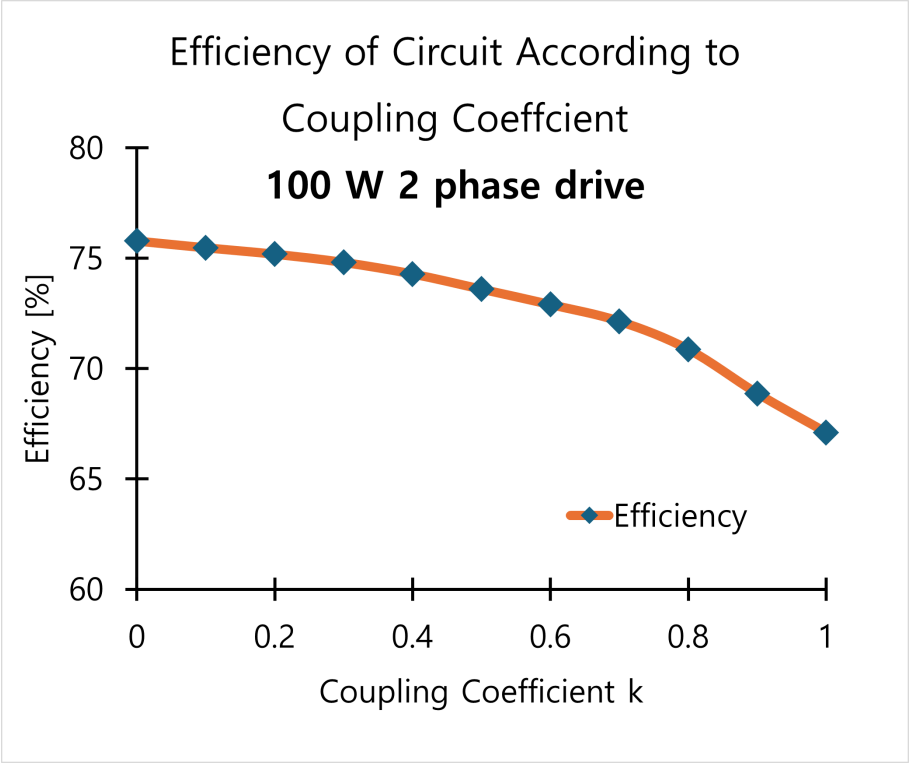


Figure 5.17: Circuit efficiency with respect to $k_{N_{12}N_{34},N_{56}N_{78}}$.

5.4 Experimental Verification

The prototype two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer was tested in different load conditions from 10 W to 240 W. For this study, the circuit was tested at 100 V V_{rms} v_{in} input voltage and 48 V output voltage V_O, with 100 kHz switching frequency f_s. The experimental setup is shown in Figure **??**.

Since no control was present in this circuit, the duty had to be changed for each load level, and the following formula was used [11]. The amplitude of the input voltage is used.

$$D = \frac{2}{v_{in}}\sqrt{\frac{0.5 L_m P_O}{T_s}} \tag{5.7}$$

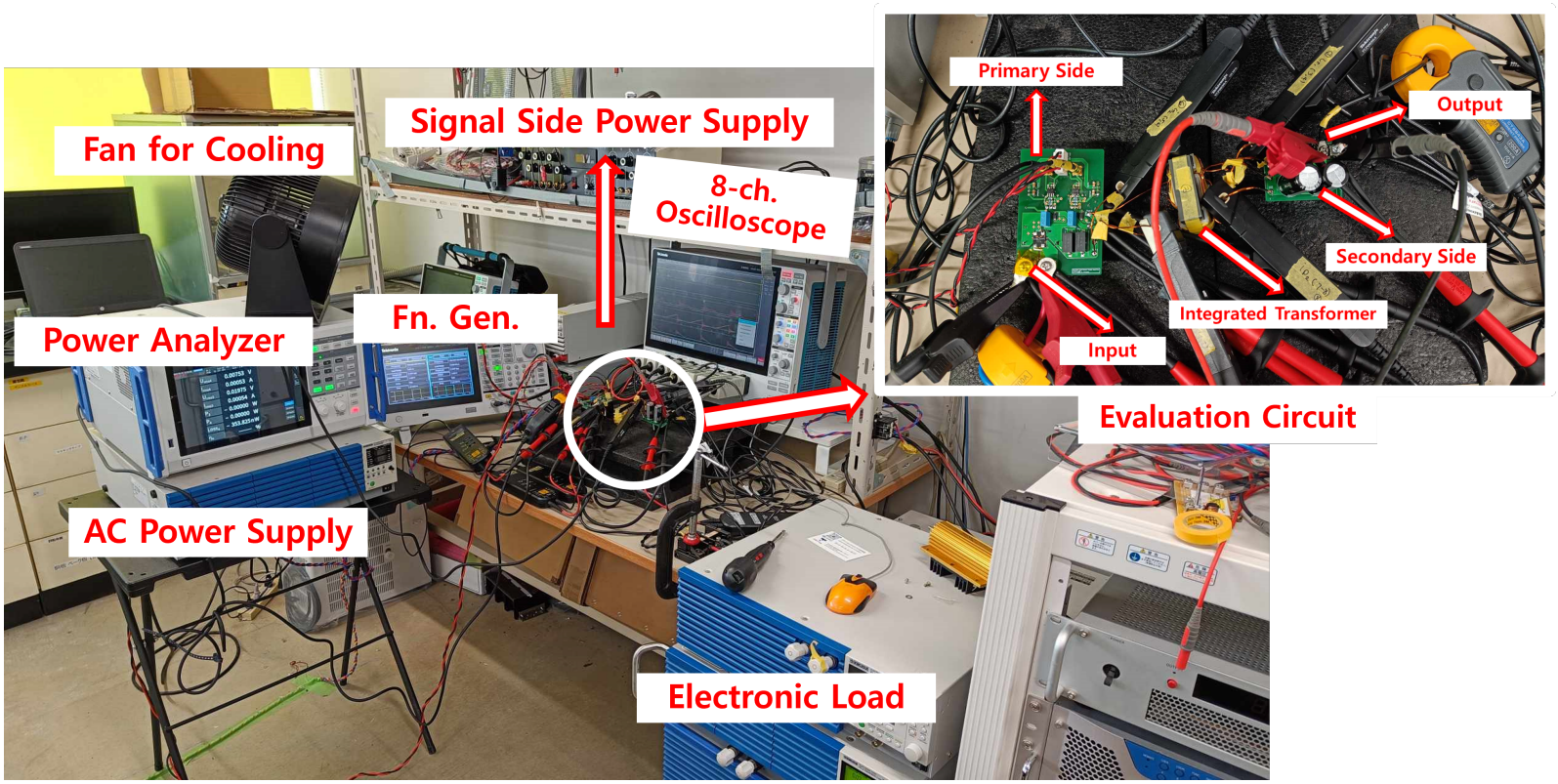
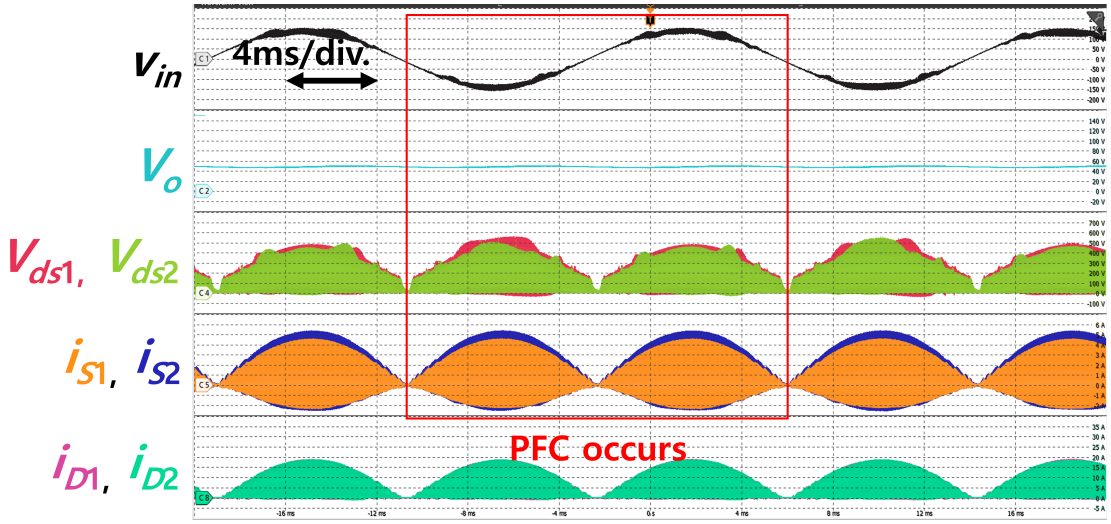


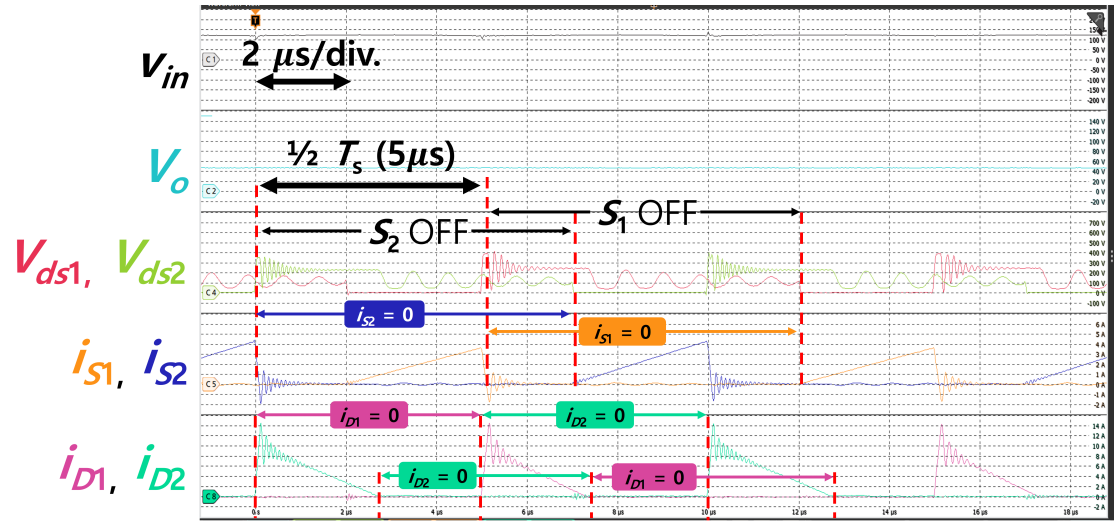
Figure 5.18: Experimental setup for the verification of operation of the proposed non-coupled integrated transformer.

5.4.1 Two-phase AC Operation

Figure **??** shows the 100 W experimental waveforms at 4 ms/div. from the two-phase interleaved flyback rectifier with the proposed transformer, demonstrating power factor correction. Figure **??** presents 2 μs/div. waveforms, with the off-times of S_1 and S_2 indicated. The current i_{S_1} is zero during the off-time of S_1 and i_{S_2} is zero during the off-time of S_2. No current is induced in either switch except during the resonance between the leakage inductance and the parasitic capacitance of the switch. During the inactive period of the transformer, for example, between the start of the off-time of S_1 and the start of the off-time of S_2, the secondary diode side current is zero. Moreover, after i_{D1} and i_{D2} reached zero when S_1 and S_2 switched off, the currents remained at zero. Lastly, a linear increase and decrease of i_{S_1}, i_{S_2}, i_{D_1}, and i_{D2} was observed throughout the circuit operation. Therefore, from experiment, the two flyback converters N_{12}, N_{34}, S_1, D_1, C_O, V_O and N_{56}, N_{78}, S_2, D_2, C_O, V_O can operate independently without magnetic coupling interference, as coupling was eliminated between the flyback transformers.



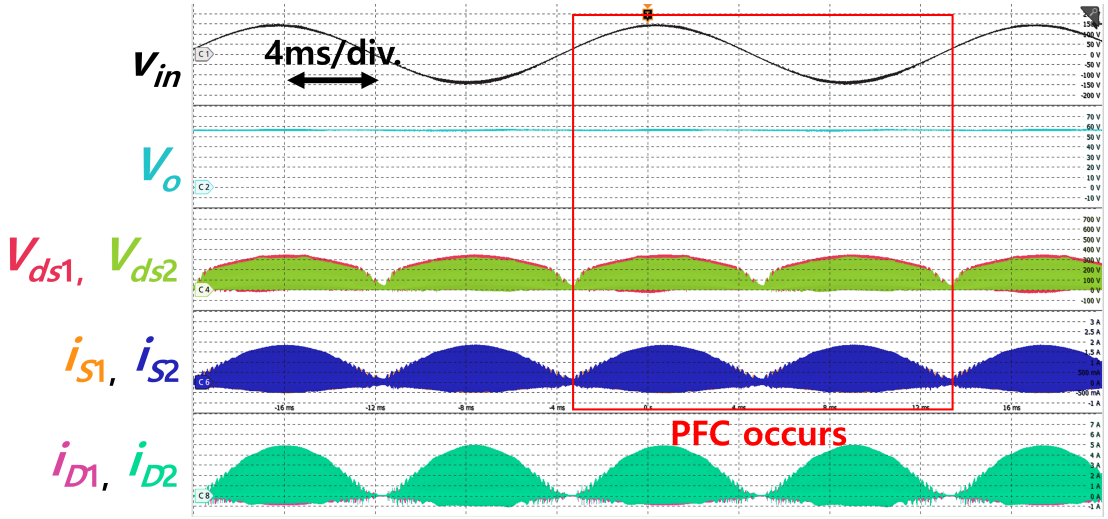
(a) Zoomed-out experimental waveforms



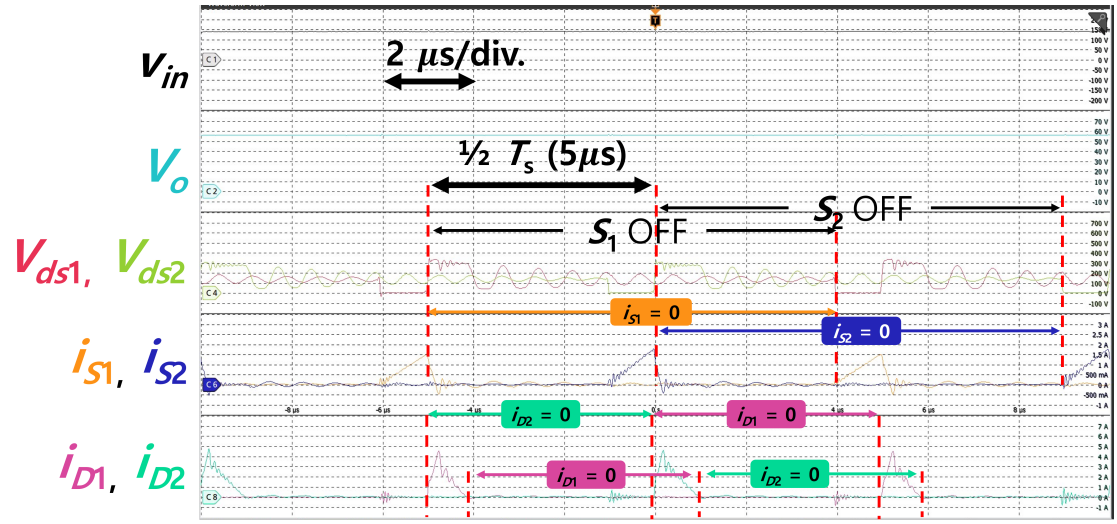
(b) Zoomed-in experimental waveforms.

Figure 5.19: Experimental waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 100 W, two-phase AC drive. Here, v_{in}: 50 V/div.; V_o: 20 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 1 A/div., i_{D_1}, i_{D_2}: 5 A/div.; v_{ds} stands for drain-source voltage.

The same phenomenon can be seen at low (10 W) and high loads (240 W), which therefore implies that the transformer can operate separately in two phases at any load with out any magnetic interference. The results are shown in Figure **??** and **??** respectively. However, as clearly seen from the result at 240 W (Figure **??**), there are are a few points of improvement for the current state of the converter and transformer. For instance, the peak values of the switch currents are not equal for each phase, which therefore means that the unequal value of magnetic inductances of the transformers (see Table **??**) affect the operation of the circuit. Moreover, from examining the waveforms of i_D, the discharge time has also been affected. For a set peak current and constant output voltage, a change in L_m will affect the discharge time. A bigger magnetizing inductance means a longer discharge time. This can be solved by ensuring precise manufacturing and core selection for the transformer, such that the magnetizing inductances of the two transformers can be easily set to be equal to each other, and such that the assumptions set in Chapter 3 regarding the core cross-sectional areas can be satisfied.

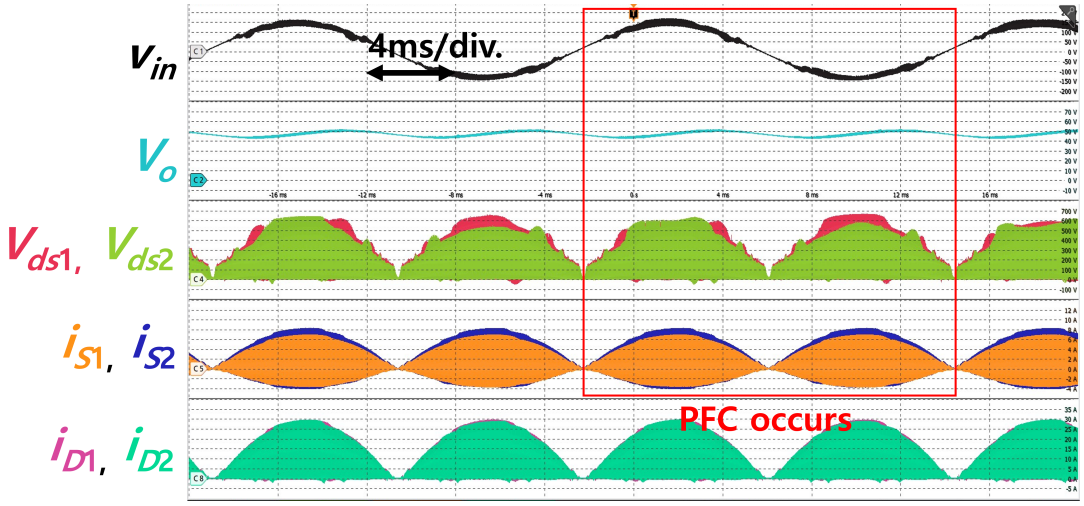


(a) Zoomed-out experimental waveforms.

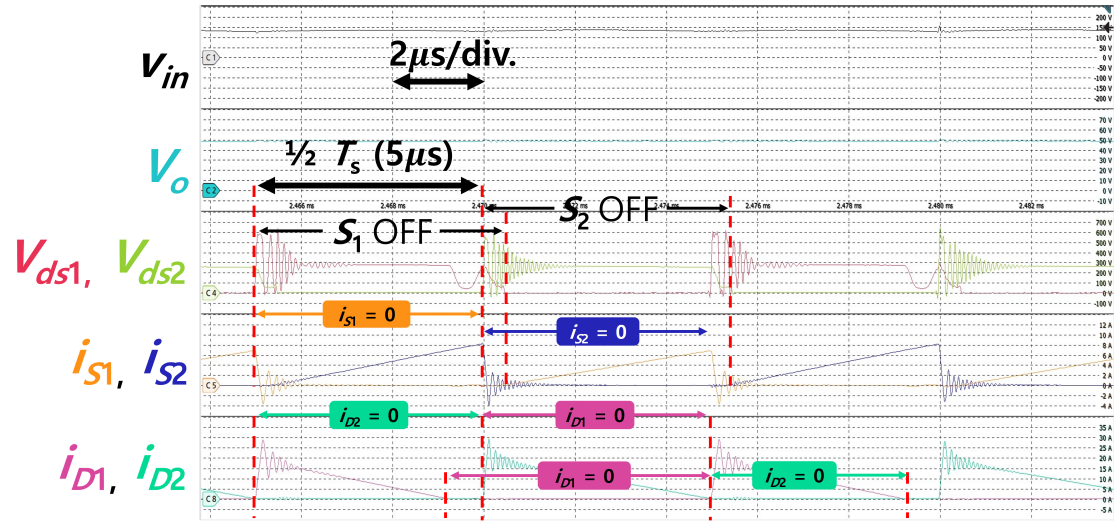


(b) Zoomed-in experimental waveforms.

Figure 5.20: Experimental waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 10 W, two-phase AC drive. Here, v_{in}: 50 V/div.; V_o: 10 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 500 mA/div., i_{D_1}, i_{D_2}: 1 A/div.; v_{ds} stands for drain-source voltage.



(a) Zoomed-out experimental waveforms.

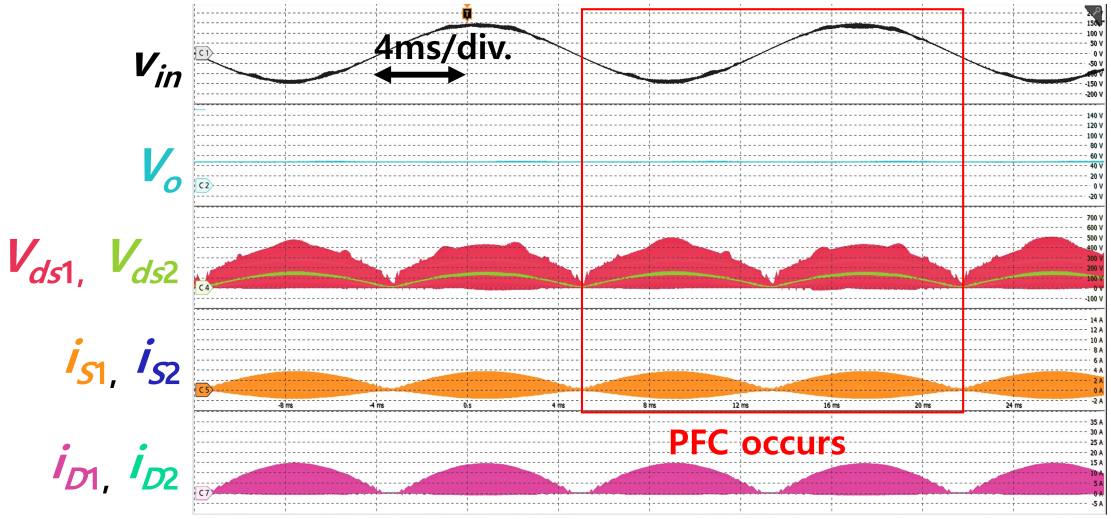


(b) Zoomed-in experimental waveforms.

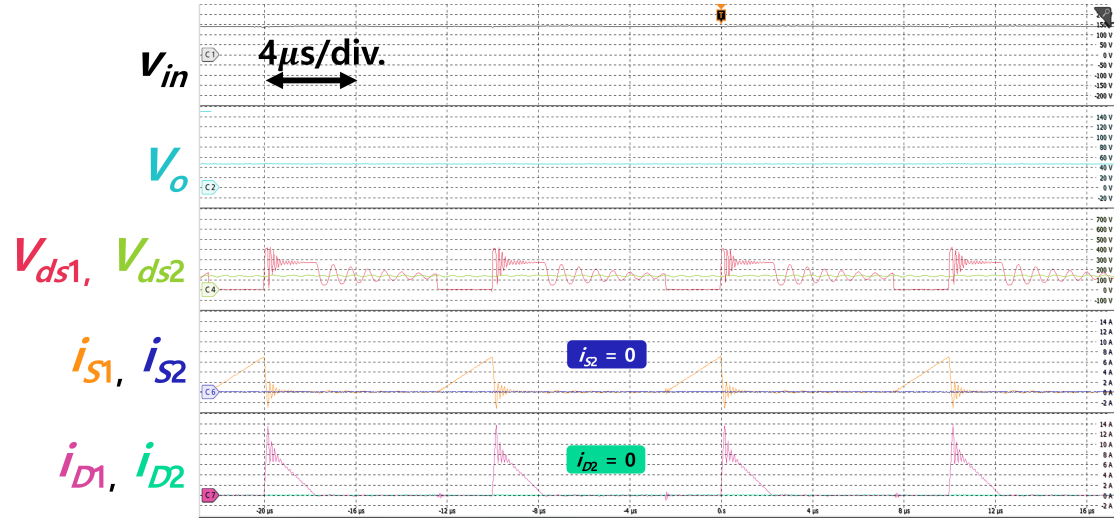
Figure 5.21: Experimental waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 240 W, two-phase AC drive. Here, v_{in}: 50 V/div.; V_o: 10 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 2 A/div., i_{D_1}, i_{D_2}: 5 A/div.; v_{ds} stands for drain-source voltage.

5.4.2 Single-phase Operation

Single-phase drive at lower power is also feasible with this transformer. Figure **??** shows the 32.5 W, single-phase drive of S_1, 4 ms/div. waveforms two-phase interleaved flyback rectifier with the proposed transformer, demonstrating power factor correction. Figure **??** presents 2 μs/div. waveforms. The currents i_{S_2} and i_{D_2} are zero all throughout the circuit operation. This then confirms that only the flyback converter N_{12}, N_{34}, S_1, D_1, C_O, and V_O are operating. There is a voltage seen at V_{DS_2} since the two flyback converters are in parallel, although the switch is kept off. Therefore, from simulation, non-influence of the inactive transformer to the active one is confirmed due to the elimination of coupling between the transformers.



(a) Zoomed-out experimental waveforms.



(b) Zoomed-in experimental waveforms.

Figure 5.22: Experimental waveforms of a two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer at 32.5 W, one-phase AC drive of S_1. Here, v_{in}: 50 V/div.; V_o: 20 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 2 A/div., i_{D_1}, i_{D_2}: 2 A/div.; v_{ds} stands for drain-source voltage.

5.4.3 Converter Efficiency

The experimental efficiency has been obtained and the results are shown in Figure **??**. Converter efficiency continuously increases with load variation. Low efficiency is expected for light loads. At the highest load (240 W), the efficiency is around 85%. To further improve the efficiency, better components and circuit design can be used to lessen component losses. Moreover, control can also be added. Lastly, as had been explained, loss analysis is going to be performed to understand more the behavior of the transformer according to the circuit specifications and be able to optimize the design of the transformer more.

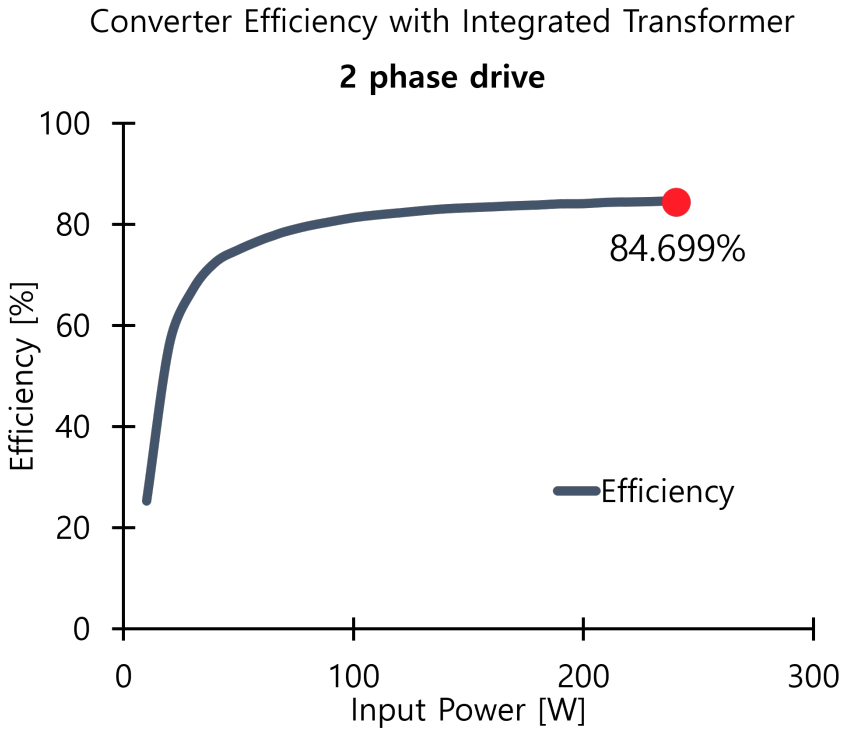


Figure 5.23: Experimental efficiency of the prototype two-phase interleaved flyback PFC rectifier with the proposed non-coupled integrated transformer (2-phase drive).

Chapter 6

Summary and Conclusion

In this paper, an integrated transformer was successfully designed for a two-phase interleaved flyback rectifier for USB PD 3.1 applications. The proposed winding structure for the integrated transformer prevents coupling between the two flyback transformers, allowing the two flyback converters to operate independently in each phase without magnetic interference. A very low coupling coefficient of 0.00053 was achieved with the prototype. Independent flyback circuit operation without magnetic interference due to coupling was verified through simulation and experiment using the designed transformer and an evaluation circuit up to 240 W. The proposed transformer is able to operate well in both single-phase and two-phase drive. Because of the proposed design, two transformers are integrated into one component, enabling the downsizing of magnetic components in the circuit. The extent of miniaturization was found to be about 19% compared to the conventional transformer. The two-phase interleaved flyback PFC rectifier with the proposed transformer then becomes an acceptable alternative to the circuits shown in the introduction of this paper.

In addition, a thorough operational analysis of the circuit with the proposed transformer has been performed, and important parameters regarding the transformer for the design consideration were analyzed and defined. A design method

and core selection method through the area product method has been demonstrated and outlined in this paper. Many different simulations have been performed too to understand fully the phenomena behind the circuit and the proposed non-coupled integrated transformer.

A main recommendation for the study is to make the magnetizing inductances of the two transformers equal, to conform to the theoretical analysis and ensure proper operation of the converter. Although, it is evident that the transformers were able to operate independently of each other across the two phases, this inequality can damage the circuit components and can affect efficiency as well. The magnetic model used in the simulation must also be corrected too in terms of the magnetizing inductances to make efficiency measurements more accurate.

In the next phase of this study, a thorough loss analysis should be performed for both the circuit and the proposed transformer. The efficiency curve in simulations performed with the conventional and proposed transformers showed very close efficiency values between the two. Although this is expected in reality, it is important to note that the core loss model for the special operation of this transformer is yet to be proposed through FEM to confirm this hypothesis. As this study demonstrated a proof of concept, the next target should be to enhance the conversion efficiency, so that it could potentially be applied in technology today. This can involve selecting components with better performance in terms of losses and further optimizing the transformer design. Control can be added also to the circuit.

The next phase of the study looks promising, especially since the transformer can be further optimized in terms of volume and geometry. This would allow the circuit to be further miniaturized, reducing cost and enhancing affordability and capability in USB PD 3.1.

This research will be presented at the

2024 Energy Conversion Congress and Exposition (ECCE 2024)

at Phoenix, Arizona, USA

October 20 - 24, 2024

Design Method of an Integrated Transformer for a Two-Phase Interleaved Flyback PFC Rectifier

Haruki Hirasawa, **Karl Raymond Dela Cruz Roque**, Sihoon Choi, Yu Yonezawa, Jun Imaoka, Masayoshi Yamamoto

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